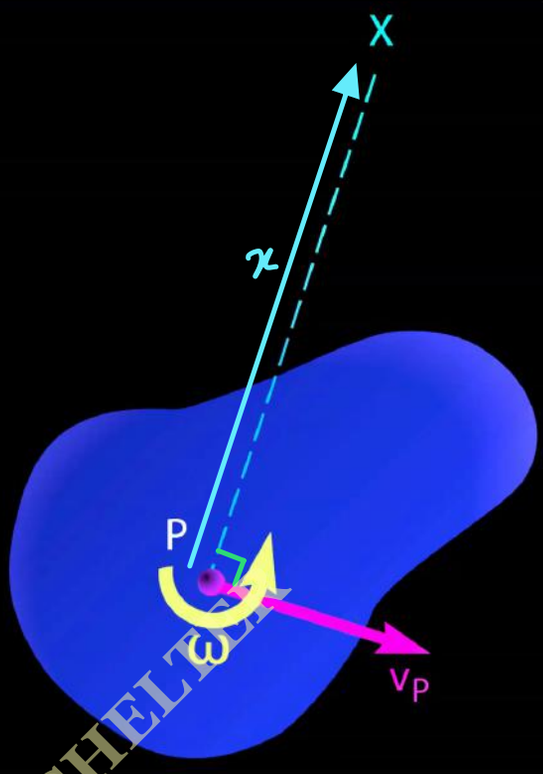
Instantaneous Axis Of Rotation:

Set of points on a rigid body or its rigid extension which have zero instantaneous velocity form the instantaneous axis of rotation. To visualize rigid extension, imagine the rigid body to be embedded inside an infinite massless sack which rotates as the rigid body rotates.



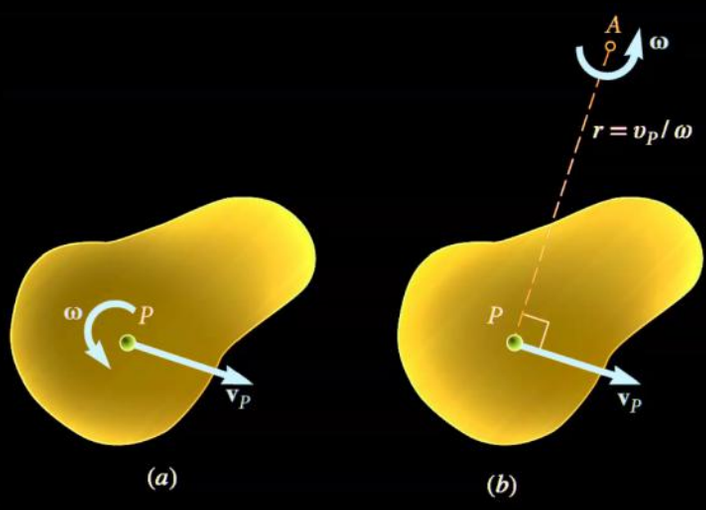
Proof Of Existence Of Instantaneous Axis:

Imagine a rigid body in plane motion (parallel to the plane of the page), now imagine an axis $P_\perp$ to the plane of the page moving with an instantaneous velocity V_P. Now recall that in a rigid body every point appears to rotate about every other point with same angular speed (say ω). Now consider a point X at a distance x from axis P on a line perpendicular to P. By superposition, velocity of this point $V_x = V_P + V_{xP} = V_P - \omega x$. Now since the extension of the rigid body is infinite therefore we will be able to locate a point X such that $V_P - \omega x = 0$, infact for $x = V_P/\omega$ we can see that velocity of X is zero (actually similarly velocity of all points on axis passing through X perpendicular to the plane of the page must be zero. Thus we have not only proved the existence of instantaneous Axis (A) but also found its location Q.E.D.

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Significance Of Instantaneous Axis: (A)

In a rigid body every point appears to rotate about every other point with same angular velocity, so from instantaneous axis frame, entire rigid body appears to rotate about the instantaneous axis itself. Thus if a particle of mass m_i is located



at a distance r_i from the instantaneous axis, its velocity is ωr_i. (Just like a fixed axis case), its KE is $\frac{1}{2} m_i (\omega r_i)^2$ (Just like fixed axis case), its angular momentum is $m_i r_i^2 \omega$ about instantaneous axis (also just like fixed axis case). These are also the values from ground frame because velocity of point of instantaneous axis is zero from ground frame. Thus: Instantaneous axis is as good as a permanent axis for purpose of calculating velocities, Kinetic Energies and angular momentum of the rigid body.

→ Note that while velocity of instantaneous axis is zero, its acceleration may not be zero. Thus acceleration of any point will be equal to acceleration of the instantaneous axis + acceleration relative to the instantaneous axis.

→ Instantaneous axis is not as good as permanent axis for purpose of finding accelerations of the particles

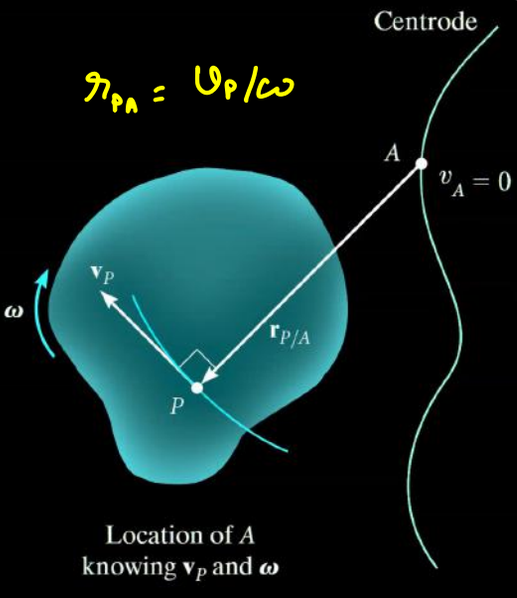
→ Thus KE of a body : $KE = \frac{1}{2} I_A \omega^2$

→ Angular momentum of a rigid body about A $L_A = I_A \omega$

→ Velocity of Arbitrary point i : $v_i = \omega \times r_{iA}$

Locating A Knowing ω And Velocity Of A Point :

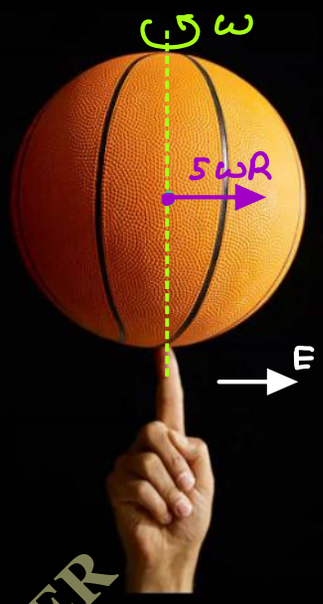
If we know the velocity of any point P and angular velocity of the rigid body, we draw a perpendicular to the velocity vector from point P and mark an intercept at a distance v_P/ω to get the location of A. (In 2D motion sometimes A is also called instantaneous centre of rotation and its locus is called centrode). Why this method works should be readily clear from the proof of existence of A.



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EXAMPLE:

A Juggler is spinning a basketball of mass M and radius R about a vertical axis while skating with a speed of $5\omega R$ to the east (a) Where is the instantaneous axis of the ball as seen from ground



(b) Where is the instantaneous axis from Juggler's frame

(c) What is the maximum speed of a particle on the ball

(d) What is the kinetic Energy of the ball

(e) What is the angular momentum of the ball about its instantaneous axis (as seen from ground).

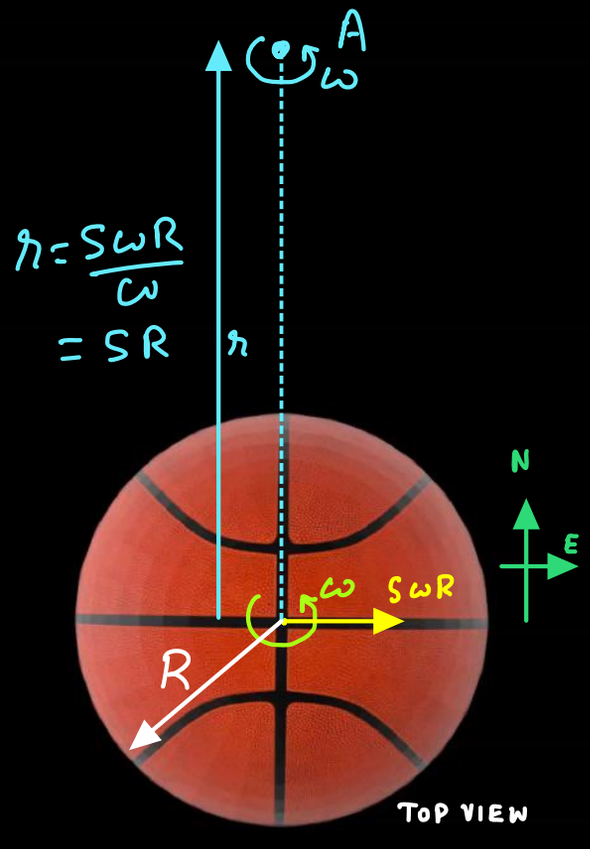
(f) What is the speed of westernmost point of the ball.

(g) Find KE of the ball using the formula for system of particles i.e $KE = K_0 + K_{cm}$ (Recall from chapter on centre of mass).

Treat basketball as a thin spherical shell.

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Solution: For ease of visualisation we should draw the rigid body in a view in which axis is perpendicular to the page. Irrespective of the shape or size of the body, we can always visualise it as an infinite body (imaginary rigid extension). Now centre of the ball moves with linear speed $5\omega R$ while ball rotates with ω. Thus instantaneous axis must be at a distance of $r = 5\omega R/\omega = 5R$ from centre of the ball (see figure).



(a) Instantaneous axis from ground is located at a distance $5R$ from the centre towards north

(b) From skater's frame ball is in fixed axis rotation thus instantaneous axis is the vertical passing through centre of

the ball.

(c) $v_i = \omega r_i$. Thus maximum speed will occur for farthest point from instantaneous axis (i.e southernmost point). Thus,
$v_{max} = \omega \times 6R = 6\omega R$ Ans. (Alternatively, $5\omega R + \omega R$ using CM)

(d) $KE = \frac{1}{2} I_A \omega^2 = \frac{1}{2}\left(\frac{2}{3}MR^2 + M(5R)^2\right)\omega^2 = \frac{77}{6}MR^2\omega^2$ Ans.

(e) $L_A = I_A \omega = \frac{77}{3}\omega$ Ans.

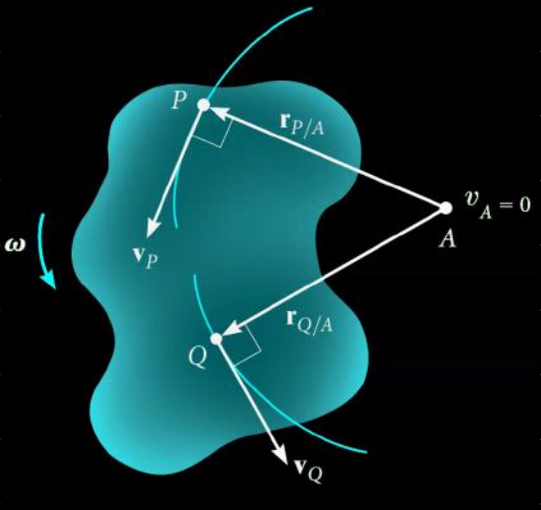
(f) Distance of westernmost point is $\sqrt{(5R)^2 + R^2} = \sqrt{26}\,R$
$\Rightarrow$ its speed is $\sqrt{26}\,\omega R$ Ans.
(Can you see that it is directed towards $\tan^{-1}\frac{1}{5}$ south of east? Try to visualise considering that every point's velocity is $\perp$ to line joining it with A)

(g) $K = K_0 + K_{CM} = \frac{1}{2}M(5\omega R)^2 + \frac{1}{2}\times\frac{2}{3}MR^2\omega^2 = \frac{77}{6}MR^2\omega^2$

(We get the same result as above).

Locating Instantaneous Axis Knowing Directions Of Velocities Of Two Points:

In the previous section we saw that A lies on a line perpendicular to the velocity at a given point. This must be true for any point on the rigid body. Thus if directions of velocities of two points on a rigid body is known, we can draw perpendiculars to the velocity directions at both points and the intersection gives us the location of instantaneous axis. (See adjacent figure).

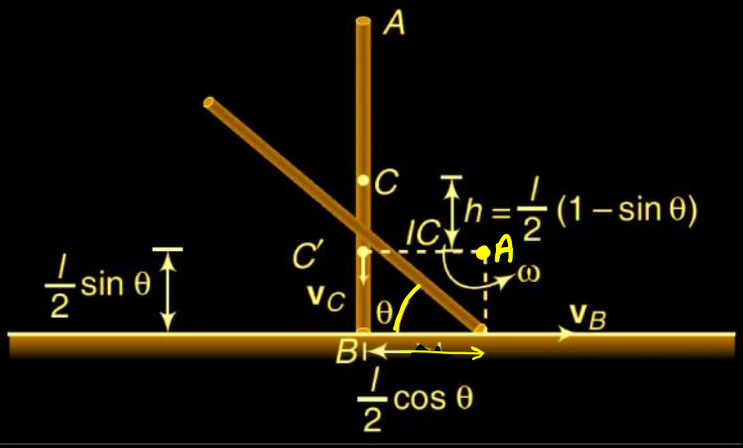


Location of A knowing the directions of $\mathbf{v}_P$ and $\mathbf{v}_Q$

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EXAMPLE :

A rod of mass m and length l begins to slide from a vertical position on a frictionless surface after being released from rest. (a) Find the angular velocity of the rod when it makes an angle θ with horizontal (b) How will you find its angular acceleration as a function of θ (c) How will you find a_{cm} (d) How will you find Normal reaction from ground at this instant.



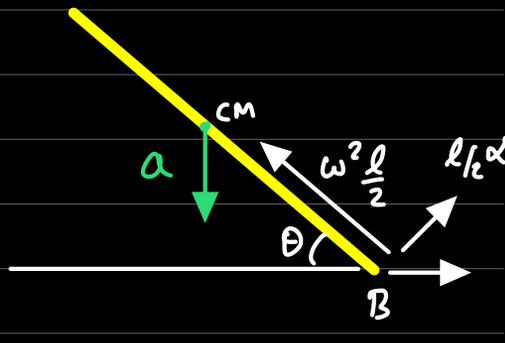
Solution: Here since there is no external force in x direction, CM must accelerate vertically, also point B must move horizontally. Drawing $\perp$ to the velocities at these points, A is marked. Now by conservation of mechanical energy:

$$mg\frac{l}{2}(1-\sin\theta) = \frac{1}{2}I_A\omega^2 = \frac{1}{2}m\left(\frac{l^2}{12} + \left(\frac{l}{2}\cos\theta\right)^2\right)\omega^2$$

$$\Rightarrow \quad \omega = \left[\frac{12g(1-\sin\theta)}{l(1+3\cos^2\theta)}\right]^{1/2} \quad \text{Ans (a).}$$

(b) Now to find α, we can use, $\alpha = \frac{1}{2}\frac{d}{d\theta}(\omega^2)$

(c) We know that vertical acceleration of point B is zero. From CM frame B has centripetal acceleration $\omega^2\frac{l}{2}$ and tangential $\alpha\, l/2$. Now using constraint, vertical acceleration of $B = 0$:



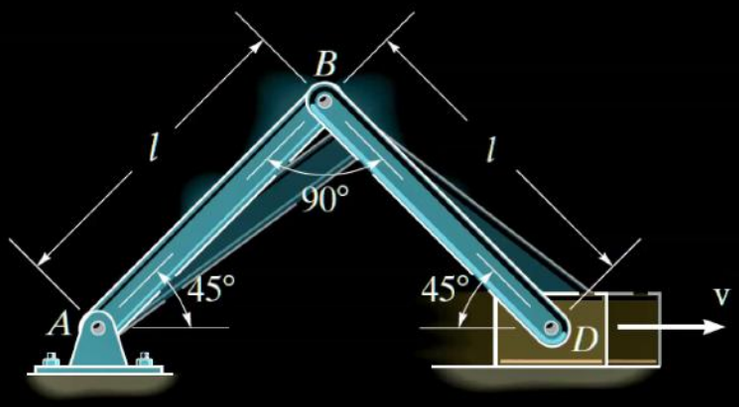
$$a - \omega^2\frac{l}{2}\sin\theta - \alpha\frac{l}{2}\cos\theta = 0 \quad \#a \text{ Ans (c).}$$

(d) Now $Mg - N = Ma$ $\#N$ Ans (d).

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EXAMPLE:

For the shown mechanism, find the angular velocities of the rods in terms of v and l.

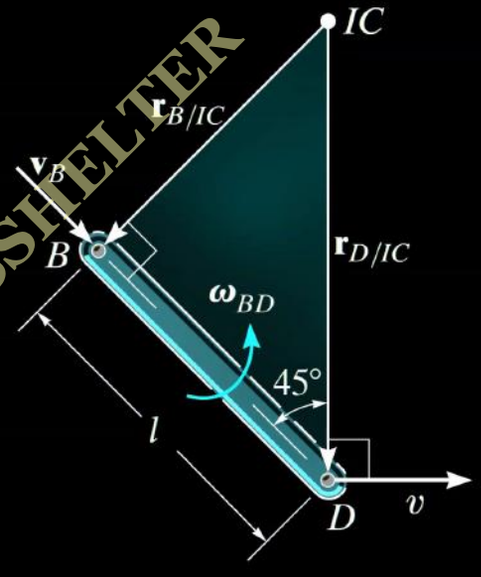


Solution: For rod AB, instantaneous axis is A (Actually permanent axis which is always an instantaneous axis). Thus motion of B is along ⊥ to AB i.e along BD. Also D is moving horizontally. Drawing ⊥ from velocities of D and B we locate instantaneous centre of rotation (IC). Thus $\omega_{BD} = \dfrac{v}{r_{D/IC}}$

$$\Rightarrow \omega_{BD} = \frac{v}{l\sqrt{2}} \circlearrowright, \quad \text{Now velocity of B:}$$

$$v_B = \omega_{BD} \times r_{B/IC} = \omega l = v/\sqrt{2}.$$

Now $\omega_{AB} = v_B/l = (v/\sqrt{2}l) \circlearrowleft$ ans.



EXAMPLE: A rod begins to slip on a frictionless surface from a vertical position. (a) How will you find ω and α in terms of θ

(b) How will you find instantaneous speeds of A & C.

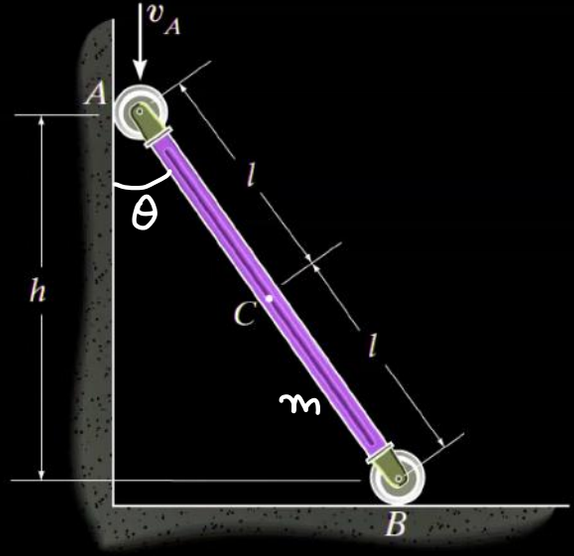
(c) What is the locus of C

(d) How will you find the KE of the rod.

(e) How will you find normal reaction from the wall

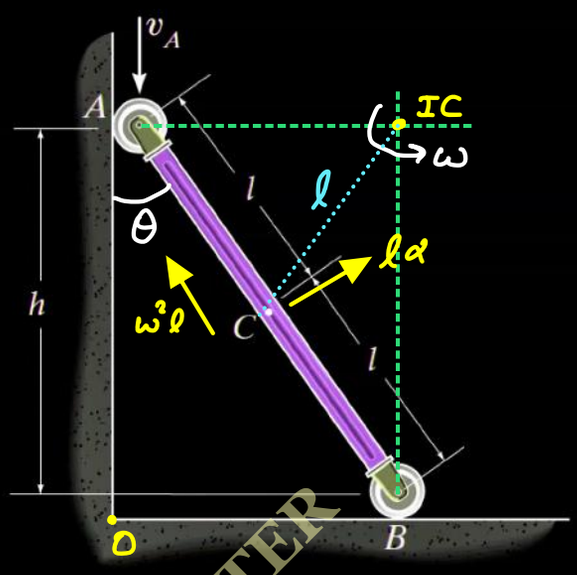
(f) The rod loses contact with wall at some θ, how will you find that angle.

Ignore the mass and dimensions of rollers.



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Solution: (a) Instantaneous axis is located by drawing ⊥ to velocities at A and B.



Now by COME:

$$mgl(1-\cos\theta) = \frac{1}{2} m\left(\frac{(2l)^2}{12} + l^2\right)\omega^2$$

Using this we can solve for ω.

Now $\alpha = \frac{1}{2}\frac{d}{d\theta}(\omega^2)$

(b) Now $v_c = \omega l$ and $v_A = 2\omega l \sin\theta$

(c) Recognize that distance of C from O remains constant ∴ its locus is a circular arc. (Until loss of contact)

(d). $KE = \frac{1}{2} I_{IC} \omega^2 = \frac{1}{2} m\left((2l)^2/12 + l^2\right)\omega^2$

(e) Acceleration of A is always vertical and from frame of A, C has centripetal acceleration $\omega^2 l$ and tangential acceleration $l\alpha$ thus horizontal acceleration of C is $a_x = l\alpha\cos\theta - \omega^2 l \sin\theta$, Now N can be found using equation $N = Ma_x$.

(f) At the time of loss of contact, $N = 0$ using this we will be able to solve for θ.

→ Note that centrode point is accelerated (its locus will be a circle of radius $2l$ about O), its vertical acceleration is same as that of A and horizontal acceleration is same as that of B [Net acceleration is double that of C ∵ its coordinates are always double that of C wrt O).

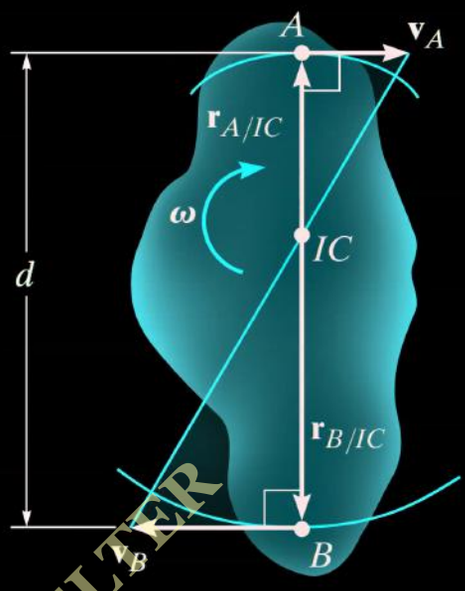
Think!: How will you find acceleration of A?

Solution: Vertical acceleration of B is zero. From frame of A, B has centripetal acceleration of $\omega^2(2l)$ and tangential acceleration $2l\alpha$ thus, using constraint of zero vertical acceleration:

$$a_A - \omega^2(2l)\cos\theta - (2l\alpha)\sin\theta = 0 \quad \# a_A.$$

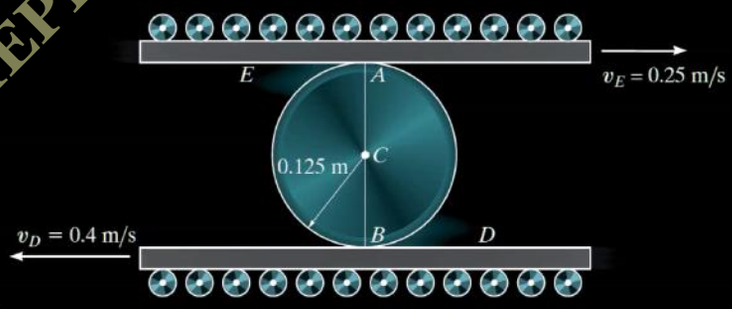
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Locating Instantaneous Axis (Know Antiparallel Velocities:



Consider the rigid body shown in the figure. Here instantaneous axis must divide the distance AB internally in the ratio of $V_A : V_B$. This is because velocities are directly proportional to distance from instantaneous axis. Alternatively we can find ω using the fact that every point appears to rotate about every point with same ω thus, $\omega = v_{AB}/\overline{AB}$, once we know ω, locating IC is easy (reduces to the case of known velocity of a point and ω). Now distance of IC from A is simply V_A/ω.

Example: For the shown figure, find (a) ω (b) location of IC (c) speed of point C.



Solution: (a) $\omega = (V_A + V_B)/2R$

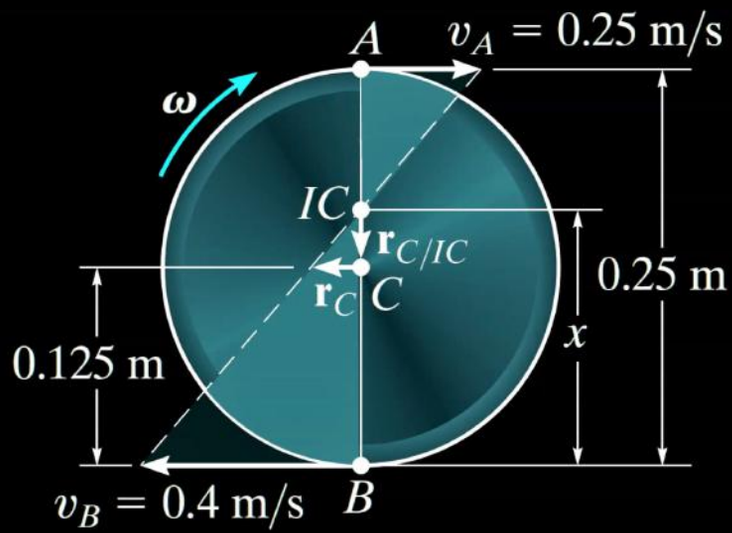
$= (0.25 + 0.4)/0.25 = 2.6$ rad s^{-1} ↻

Now location of IC from A, V_A/ω

$\Rightarrow r_{IC/A} = 0.0962$ m

Now speed of C: $v_C = \omega \times r_{C/IC}$

$= 2.6 \times (0.125 - 0.0962) = 0.0748$ m s^{-1} ←



→ Alternative we could use section formula for velocity of C i.e

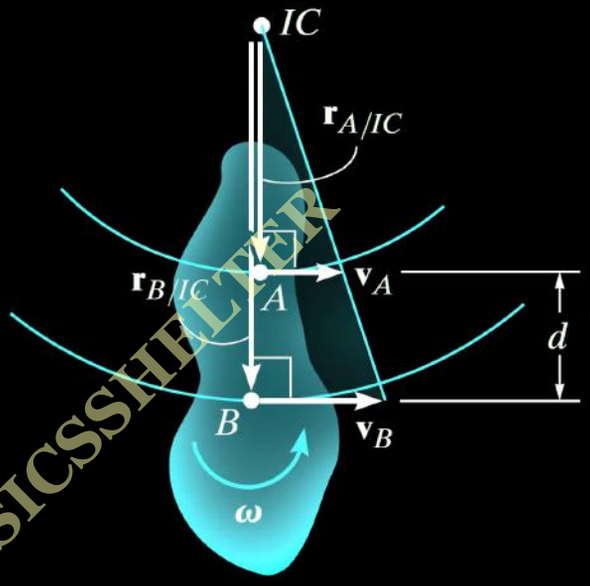
$V_C = (V_A + V_B)/2 = \dfrac{(0.4 - 0.25)}{2} = 0.075$ m s^{-1} ←

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→ Section formula can be always used to find velocities on acceleration of a point on line joining two points with known velocities and accelerations.

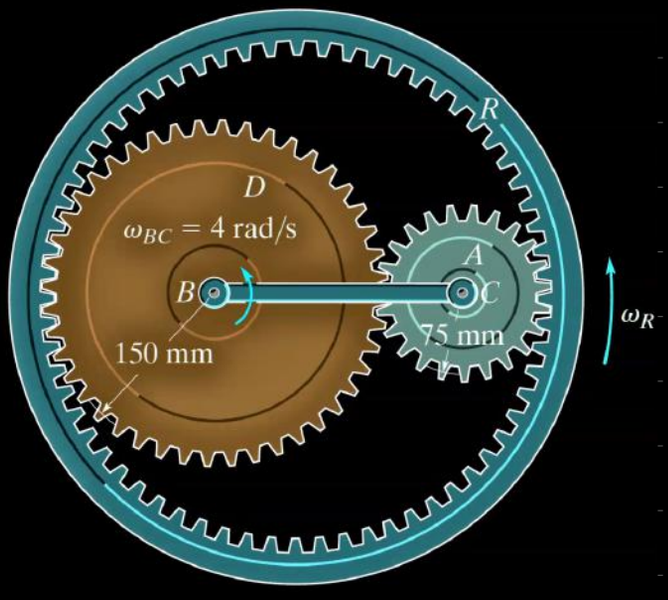
LOCATING INSTANTANEOUS AXIS (KNOWN || VELOCITIES):

Consider the adjacent figure where the velocities of points A and B are parallel to each other. Here instantaneous axis will externally divide the distance AB in ratio of their velocities,
i.e $r_{A/IC} / r_{B/IC} = v_A / v_B$. Here we



can also readily find ω as $\omega = (v_B - v_A)/d$ and now instantaneous axis can be located using, v/ω idea (first case of locating IC).

EXAMPLE: Planet gear A is pin connected to the end of link BC. If the link rotates about the fixed point B at 4 rad s⁻¹ determine the angular velocity of ring gear R. The sun gear D is fixed from rotating.



Solution: Velocity of point C:

$v_c = \omega_{BC} \times BC = 4(225) = 900 \text{ mm s}^{-1}$

The contact point between A and D is at rest (∵ no slipping) ⇒

$\omega_A = v_c / 75\text{mm} = 900 \text{ mm s}^{-1} / 75 \text{ mm} = 12 \text{ rad s}^{-1}$

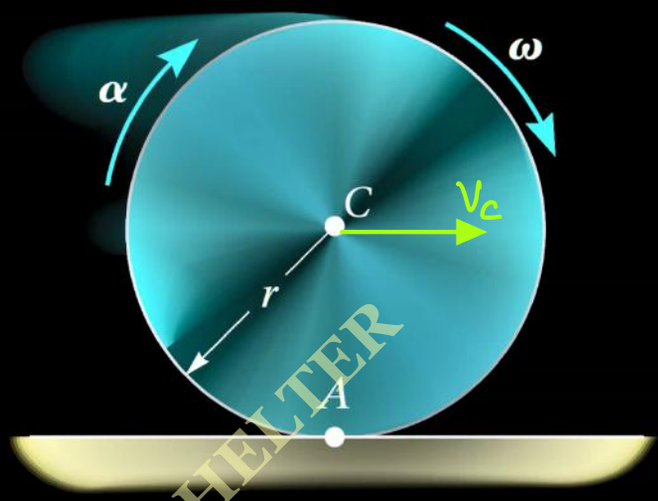
Now velocity of contact point of A and R = $v_c + \omega_A \times 75\text{mm} = 900 \text{ mm s}^{-1} + 12 \times 75 \text{ mm s}^{-1}$

⇒ Velocity of contact point of R with A = 1800 mm s^{-1}

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Now $\omega_R = \dfrac{1800\ \text{mm}^{-1} - 0}{(150\times 2 + 75\times 2)} = 4\ \text{rad s}^{-1}$ Ans.

Rolling Without Slipping:



Consider a wheel moving on a floor such that its angular velocity is ω and speed of its centre is V_c. Thus speed of the bottommost point, $\vec{V}_A = \vec{V}_c + \vec{V}_{Ac} = V_c - \omega r$.

Now recall that for no slipping between two points their velocities must be equal. Thus $V_c - \omega r = 0$ (∵ point of ground has zero velocity.) This gives us condition for pure rolling as:

$$\boxed{V_c = \omega r} \quad (1)$$

If $V_c > \omega R \Rightarrow$ Forward slipping
If $V_c < \omega R \Rightarrow$ Backward slipping.

Now let a_c be the acceleration of centre of the wheel, the differentiating (1) wrt time:

$$\boxed{a_c = \alpha r}$$

If $a_c > \alpha r \Rightarrow$ Forward slipping
If $a_c < \alpha r \Rightarrow$ Backward slip
(Assuming starting from rest).

→ Note that A is instantaneous axis since its velocity is zero.

Trajectory Of Particle On Periphery Of A Rolling Wheel (Cycloid):

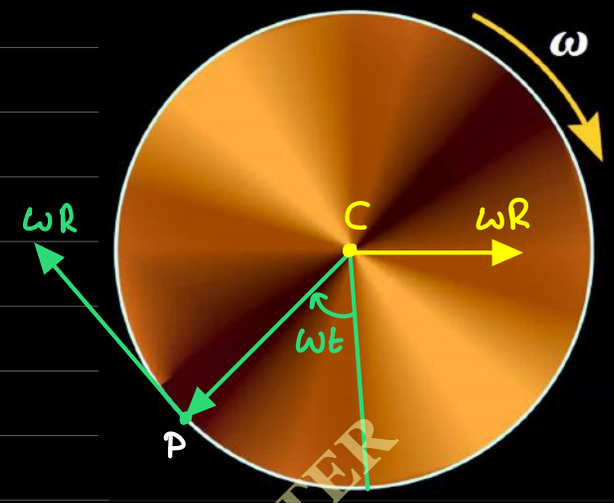


The trajectory of a particle on periphery of a rolling wheel is shown in the figure above. It is called a cycloid.

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LENGTH OF A CYCLOID IN ONE COMPLETE CYCLE:

We wish to find length of the path traced by particle on periphery of a rolling wheel in one complete cycle. (Suppose from time when bottommost particle was at A to when it is at A'). We start our clock from position A. Now in a time t, the wheel as rotated through an angle ωt. The centre moves with a speed ωR and a particle on periphery moves with a speed ωR relative to the centre. Thus its net speed is given as:



$$v = \sqrt{(\omega R)^2 + (\omega R)^2 + 2(\omega R)^2 \cos(180^\circ - \omega t)} = \sqrt{2}\omega R \sqrt{1 - \cos\omega t}$$

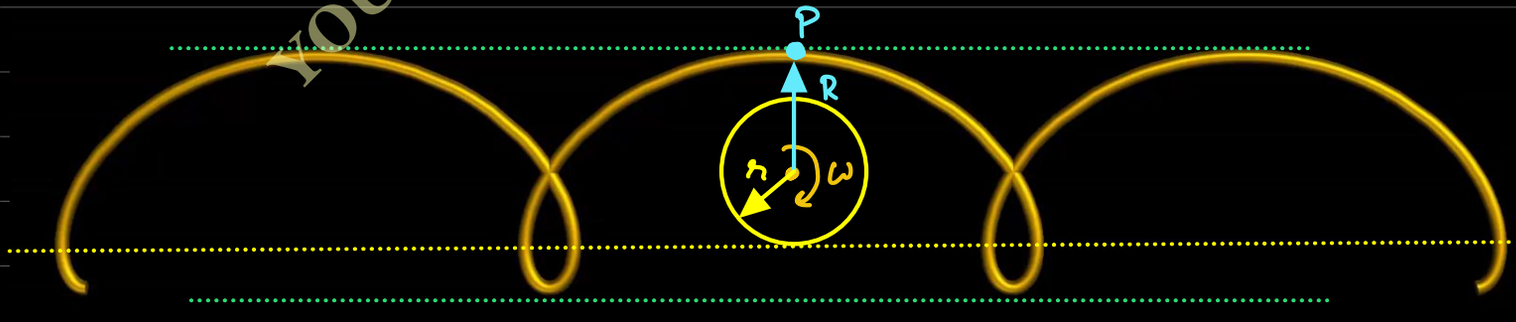
$$\Rightarrow v = 2\omega R \sin(\omega t/2).$$ Now arc length covered in time dt, $dl = v\,dt$

Thus, total arc length in one time period:

$$l = \int v\,dt = \int_0^{2\pi/\omega} 2\omega R \sin(\omega t/2)\,dt = 8R$$

Thus $\boxed{l = 8R}$ → A cycloid is also known as common trochoid.

SUPERIOR TROCHOID:



The path of a particle on imaginary extension of a rolling wheel is called a superior trochoid if the distance of such a particle from centre of the wheel is more than the radius of the wheel itself. For instance in the above figure, $R > r$.

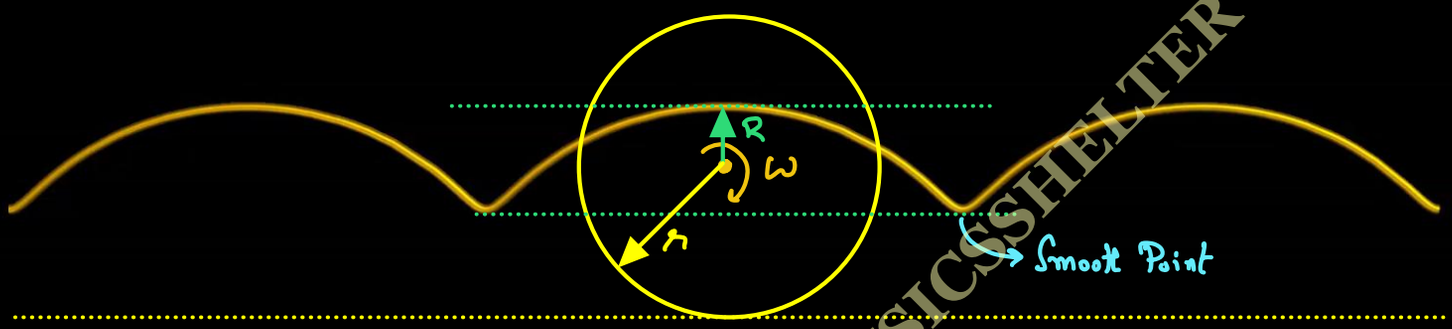
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→ A superior trochoid is a self intersecting path. The velocity of P is vertical when it is at ground level and horizontal at highest and lowest points.

→ The point P will never overshoot the upper green line (dotted) and also never undershoot the lower dotted green line.

→ Below ground level horizontal component of velocity is backward.

INFERIOR TROCHOID:



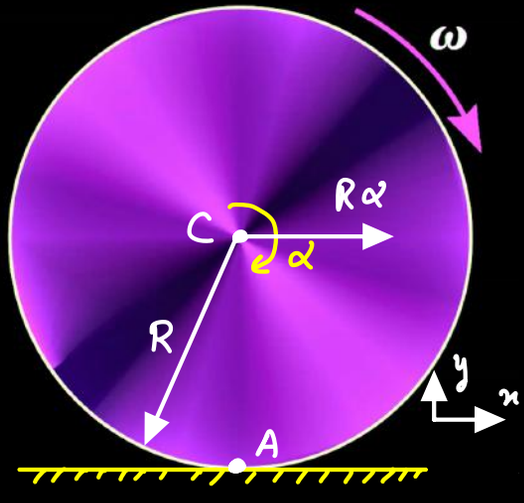
Trajectory of a particle on rolling wheel which is at a distance smaller than the radius of the wheel is called inferior trochoid.

→ The trajectory will always lie between the bounds indicated by dotted green lines.

→ The velocity of such a particle is never vertical or backward (unlike superior trochoid)

ACCELERATION OF INSTANTANEOUS AXIS OF A ROLLING WHEEL:

Consider a wheel rolling with angular acceleration α and angular velocity ω. Let A be the instantaneous axis (bottommost point) of the wheel. As seen from centre of the wheel, acceleration of A wrt C is given as $\vec{a}_{AC} = \omega^2 R\,\hat{j} - R\alpha\,\hat{i}$, Also acceleration of the centre is given as $\vec{a}_c = R\alpha\,\hat{i}$



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Now net acceleration of A : $\vec{a}_A = \vec{a}_c + \vec{a}_{Ac} = \omega^2 R\,\hat{j}$

Thus acceleration of instantaneous axis of rotation is given as:

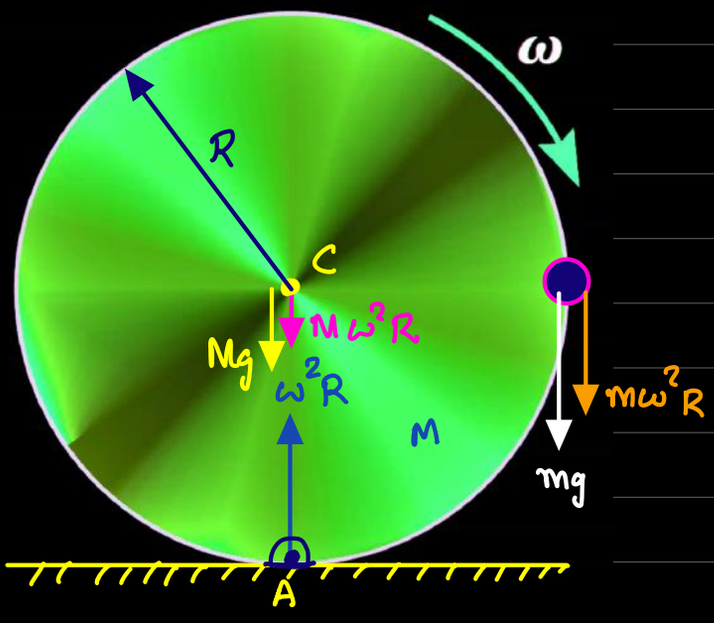
$$\boxed{\vec{a}_A = \omega^2 R\,\hat{j}}$$

→ Knowing the acceleration of instantaneous axis enables us to work from its frame (we just need to apply appropriate pseudo forces to take care of the fact that instantaneous axis is non-inertial. Once pseudo force is applied, instantaneous axis appears instantly like a fixed axis for purpose of applying $\vec{\tau} = I\vec{\alpha}$, or $\alpha = P/I_{eff}\,\omega$.

→ For a pure rolling wheel on ground, velocity of bottommost point is zero and if speed of centre of the wheel is v, that of topmost point will be $2v$

Converting A Rolling Problem To Fixed Axis Problem:

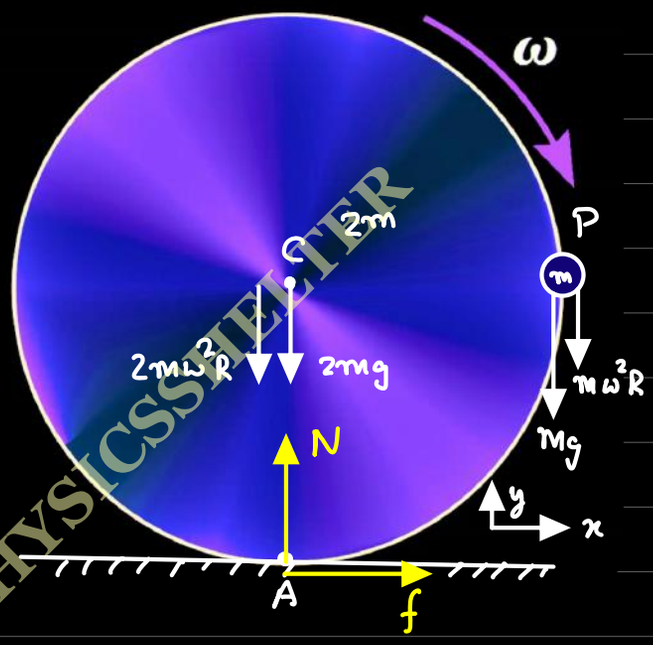
Consider a wheel of mass M rotating with an angular velocity ω instantaneously as shown in the figure. The wheel has a point mass m welded at its periphery and we want to find angular acceleration of the wheel. We know that from the frame of instantaneous axis, the entire wheel will appear to rotate around particles of instantaneous axis. However ∵ instantaneous axis is accelerated, accordingly we apply pseudo forces on the point mass and the wheel and now we can readily apply $\tau = I\alpha$ or $\alpha = P/I\omega$ about instantaneous axis.



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→ Definition of instantaneous axis depends on the frame. If we sit on any axis on the rigid body then that axis itself becomes the instantaneous axis. Just that we need to apply appropriate pseudo forces to apply $\tau = I\alpha$ and $\alpha = P/I_{PIF}\,\omega$ equations.

EXAMPLE: For the shown wheel find the angular acceleration α. The mass of the wheel is $2m$ and instantaneous angular velocity is ω. The mass of the welded particle is m and the wheel is rolling without slipping. Also make equations to find frictional force and Normal reaction.



Solution: FBD of the wheel from frame of A is shown in the figure.

Now using $\tau = I\alpha$, $(mg + m\omega^2 R)R = \left[\frac{3}{2}(2m)R^2 + m(\sqrt{2}R)^2\right]\alpha$

$\Rightarrow \quad \alpha = (g + \omega^2 R)/5R$ Ans.

Now we will used $\vec{F}_{ext} = \Sigma m_i \vec{a}_i$ to evaluate N and f. For this we need to find acceleration of m. We can either continue to use 'A' frame or we can switch to ground frame. To make these equations we need to first find acceleration of m from ground frame.

Now $\vec{a}_p = \vec{a}_c + \vec{a}_{pc} = R\alpha\,\hat{i} - R\alpha\,\hat{j} - \omega^2 R\,\hat{i}$

Thus for normal equation we can write $\Sigma F_{ext_y} = \Sigma m_i a_{y_i}$.

Thus:

$$2mg + mg - N = mR\alpha \quad \#N \quad (\text{No Pseudo forces} \because \text{Ground frame}).$$

Similarly to solve for friction we can write $\Sigma F_{ext_x} = \Sigma m_i a_{x_i}$.

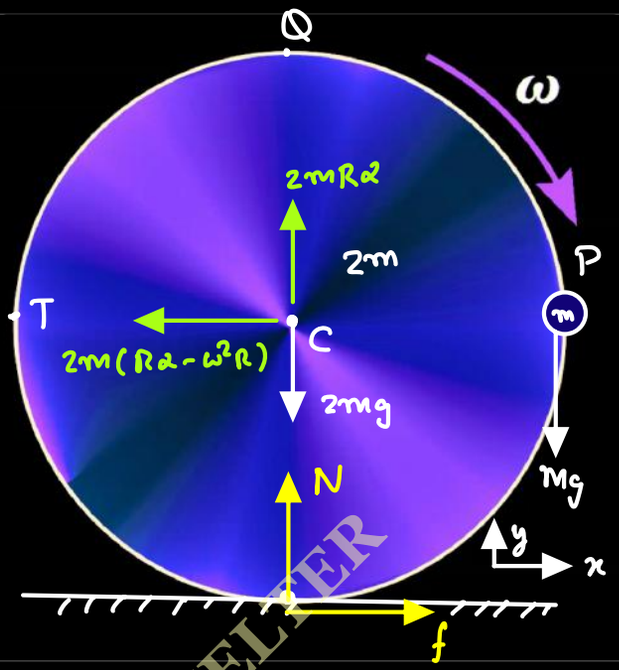
$\Rightarrow$

$$f = 2mR\alpha + m(R\alpha - \omega^2 R) \quad \#f$$

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EXAMPLE :

For the situation in previous problem, write the torque equation working from frame of P.



Solution: To work from frame of P, we will need to work out the acceleration of P first to apply appropriate pseudo forces. Now $\vec{a}_p = \vec{a}_c + \vec{a}_{pc}$ thus:

$\vec{a}_p = R\alpha\,\hat{i} - R\alpha\,\hat{j} - \omega^2 R\,\hat{i}$ i.e

$(R\alpha - \omega^2 R)\,\hat{i} - R\alpha\,\hat{j}$

Since $\vec{a}_p$ has two components of acceleration we will apply two pseudo forces on the wheel, these are shown in green colour. We also need to apply pseudo force on m itself however it will not generate any torque about P so we have not shown it. Now making torque equation about P:

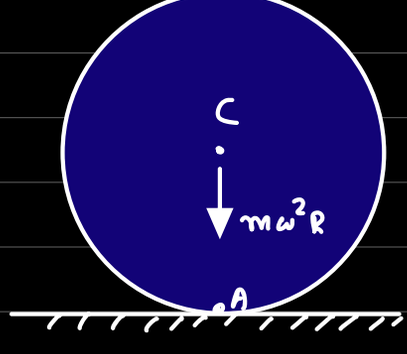
$-2mgR + NR + (2mR\alpha)R - fR = \frac{3}{2}(2m)R^2\alpha$ Ans.

→ The student is encouraged to write similar torque equations about points Q and T also.

→ Often torque equations are easy about bottommost point (∵ torque of N and f vanish), also sometimes C is a good choice for making torque equations.

TORQUE EQUATIONS FOR A "BALANCED" WHEEL :

A balanced wheel is the one for which centre of mass coincides with centre of the wheel. For such a wheel it can be readily seen that torque of pseudo force about A will vanish. So we may not worry about it while writing $\tau = I\alpha$ (unlike previous problems where the wheel was unbalanced).



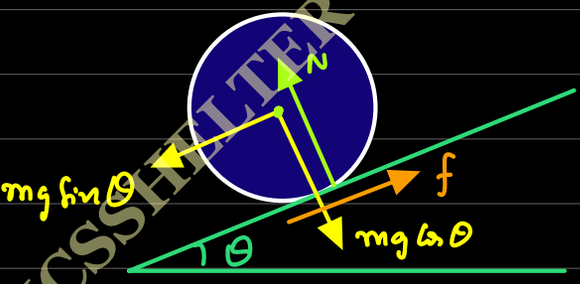
→ If we work from CM frame of the wheel, the

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pseudo forces will pass through the CM itself. Thus while writing torque equations about CM, once again we need not worry about torque of the pseudo force. Thus two good points to form equations for a balanced wheel are A and C. Accordingly:

$\boxed{\tau_{CM} = I_{CM}\,\alpha}$ and $\boxed{\tau_A = I_A\,\alpha}$

Minimum Friction Coefficient To Prevent Slipping On Incline



Consider a wheel freely rolling without slipping on an inclined plane due to gravity. Let I_{CM} be its moment of inertia about CM and I_A be its moment of inertia about A then:

$$mgR\sin\theta = I_A\,\alpha \qquad (\because \tau_A = I_A\alpha)$$

and $$fR = I_{CM}\,\alpha \qquad (\because \tau_{CM} = I_{CM}\alpha)$$

$\Rightarrow$

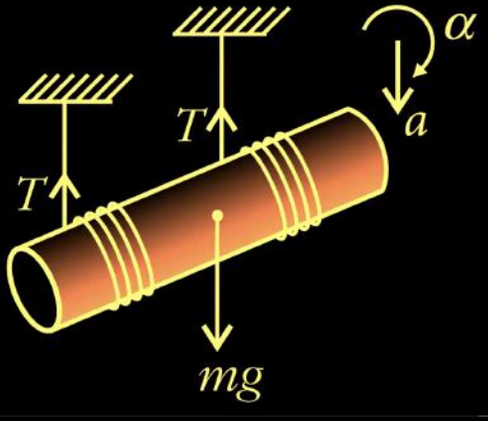
$$\frac{f}{mg\sin\theta} = \frac{I_{CM}}{I_A}$$ also $mg\cos\theta = N \Rightarrow mg\sin\theta = N\tan\theta$

$$\Rightarrow \frac{f}{N\tan\theta} = \frac{I_{CM}}{I_A} \Rightarrow \frac{f}{N} = \frac{I_{CM}}{I_A}\tan\theta$$, now for no slipping $f < \mu N \Rightarrow \mu > f/N$ thus for no slipping:

$$\boxed{\mu > \frac{I_{CM}}{I_A}\tan\theta}$$

Example: Irodov 1.253:

A uniform cylinder of mass m and radius R starts descending due to gravity. Find (a) Tension in each thread and angular acceleration (b) Time dependence of instantaneous power due to gravity.



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Solution: This is a case of pure rolling on a vertical base (thread serving as the base). Now using $\tau_A = I_A \alpha$:

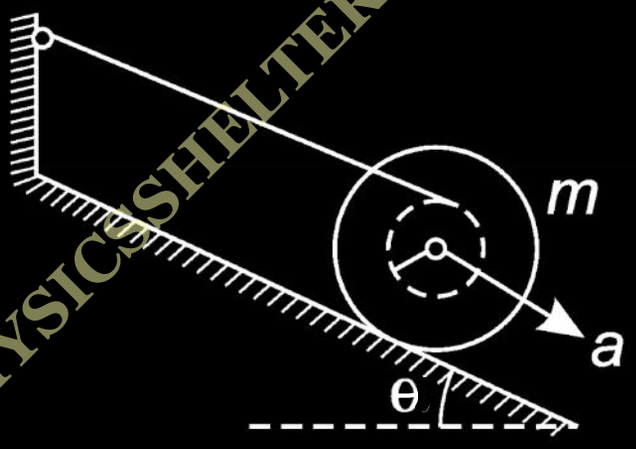
$mgR = \frac{3}{2} mR^2 \alpha \Rightarrow \alpha = 2g/3R$ ans. Now:

$mg - 2T = mR\alpha = 2mg/3 \Rightarrow T = mg/6$ ans (Each thread)

Power developed by gravity, $\tau_A \omega$ i.e $mgR \times \alpha t$ thus, power developed by gravity, $P_G = \frac{2mg^2}{3} t$ ans.

EXAMPLE: IRODOV: 1.255:

A spool with thread wound on it is placed on a smooth incline at an angle θ as shown. The free end of the thread is connected to a wall. The mass of spool is m, its moment of inertia about centroidal axis is I and radius of wound thread layer is r. Find the acceleration of the spool.



Solution: Once again this is rolling on the thread. Using $\tau_A = I_A \alpha$,

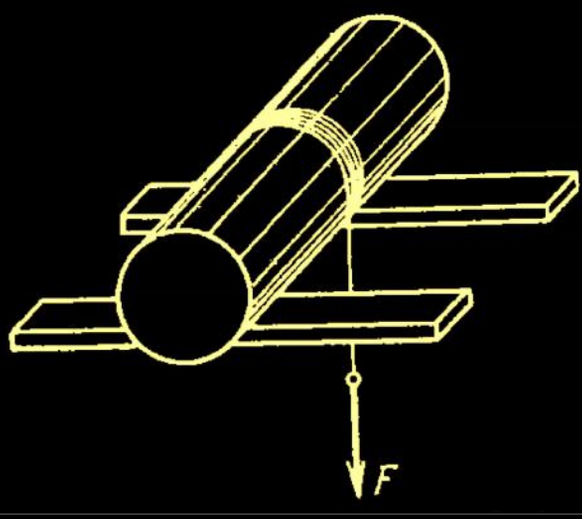
$mg \sin\theta \, r = (I + mr^2)\alpha \Rightarrow \alpha = mgr \sin\theta / (I + mr^2)$.

Now $a = r\alpha \Rightarrow a = mgr^2 \sin\theta / (I + mr^2)$

$\Rightarrow a = g \sin\theta / (1 + I/mr^2)$ ans.

EXAMPLE: IRODOV 1.256:

A uniform cylinder of mass m rests on two horizontal planks. A thread is wound on it and the hanging end is pulled with a force F as shown. If the coefficient of friction is μ, find maximum allowable value of F so that there is no slipping. What is the corresponding acceleration.



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Solution: Using $\tau_A = I_A \alpha$, $FR = \frac{3}{2} mR^2 \alpha$ ①

Using $\tau_{cm} = I_{cm} \alpha$ $-fR + FR = \frac{1}{2} mR^2 \alpha$ ②

Also $F + mg = N$ ③ and $f = mR\alpha$ ④

From ① & ④:

$$F = 3/2\, f \Rightarrow f = \frac{2}{3} F \quad ⑤$$

Using ⑤ & ⑦:

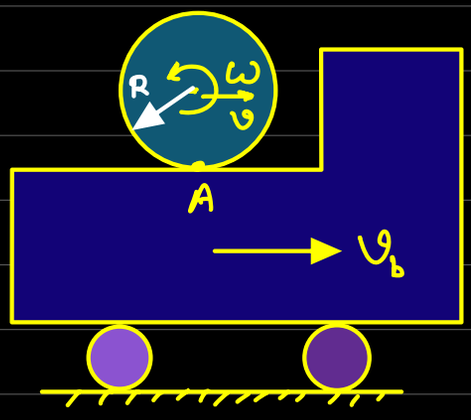
$$\frac{f}{N} = \frac{2/3\, F}{F + mg} \leq \mu \quad ⑥$$

From ⑥, $F \leq 3\mu mg/(2-3\mu)$ Ans.

Corresponding acceleration $a_{max} = f/m = \frac{2}{3} F/m = 2\mu g/(2-3\mu)$ Ans.

→ If $\mu > 2/3$, F can go even up to infinity.

Pure Rolling On A Moving Base:



When a wheel is rolling on a moving base then velocity of its centre from the frame of the base will follow the rolling condition i.e $V_{rel} = \omega R$ or

$$v - v_b = \omega R$$

(v_b: Base velocity, v: Velocity of C of the wheel).

Similarly if the base is accelerated, then for pure rolling:

$$a - a_b = R\alpha \quad (a_b: \text{Base acceleration}, a: \text{acc}^n \text{ of C of wheel}).$$

→ For problems involving rolling on a moving base, we can work in the frame of the base after applying appropriate pseudo forces on alternatively we can make equations about CM of the wheel without worrying about torques of pseudo forces (∵ they will be zero). However while writing force equations always remember to consider pseudo forces appropriate to the frame.

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EXAMPLE : IRODOV 1·258:



The arrangement shown in the figure has two identical uniform cylinders each of mass m on which two light threads are wound symmetrically. Find the tension in each thread. (There is no friction in the axles).

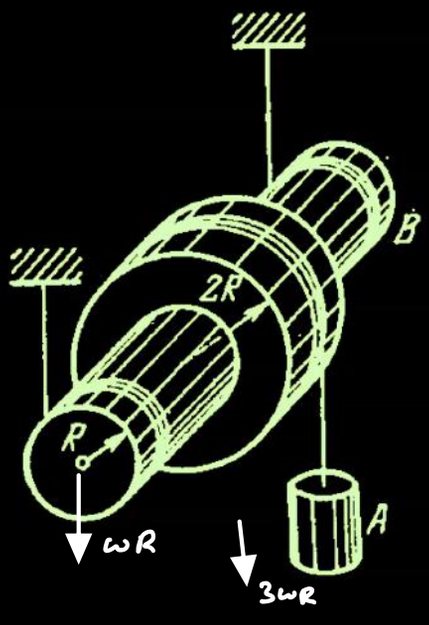
Solution: Recognize that torque of tension about axles of both cylinders has same magnitude. Thus angular accelerations will also have same magnitude (∵ I_{cm} is also identical). Now for the lower cylinder, unwinding threads form a moving accelerated base for pure rolling such that acceleration of the base is $R\alpha$. Now applying pseudo force at the centre of lower cylinder, and making torque equation $\tau_A = I_A\alpha$:

$$(mg - mR\alpha)R = \frac{3}{2}mR^2\alpha \quad (1)$$ Also:

$$mg - 2T - mR\alpha = mR\alpha \quad (2) \quad (\because F = ma \text{ from base frame}).$$

Solving (1) & (2), $T = mg/10$, $\alpha = 2g/5R$.

EXAMPLE : IRODOV : 1 259



In the shown system, the mass of the block is m while that of the stepped pulley B is M. The moment of inertia of pulley relative to its own axis is I. Find acceleration of the weight after the system is set free.

Solution: Here effective moment of Inertia about instantaneous axis is $I + MR^2 + m\times(3R)^2$

$$\Rightarrow I_{eff} = I + MR^2 + 9mR^2$$

Now $$\alpha = \frac{P}{I_{eff}\omega} = \frac{Mg\omega R + mg(3\omega R)}{(I + MR^2 + 9mR^2)\omega}$$

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$\Rightarrow \alpha = \dfrac{(M+3m)g}{(I + MR^2 + 9mR^2)}$, Now acceleration of weight $a = 3R\alpha$ Ans.

→ If $3R\alpha > g$ then the acceleration of block will be g and thread will become lose. Irodov has missed this case. (Tension in the string can't be negative obviously, so a mathematical acceleration of more than g for the block actually means that thread must have become lose (or it is pushing, which is not possible)).

EXAMPLE: IRODOV: 1·260:

A uniform solid cylinder A of mass m_1 can freely rotate about a horizontal axis fixed to a mount B of mass m_2 (Fig. 1.66). A constant horizontal force F is applied to the end K of a light thread tightly wound on the cylinder. The friction between the mount and the supporting horizontal plane is assumed to be absent. Find :

(a) the acceleration of the point K;

(b) the kinetic energy of this system t seconds after the beginning of motion.

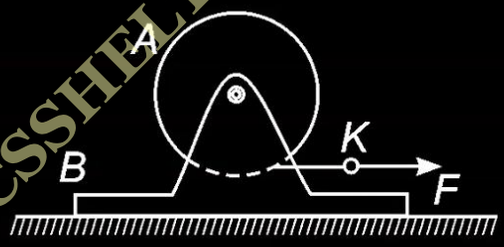


Fig. 1.66

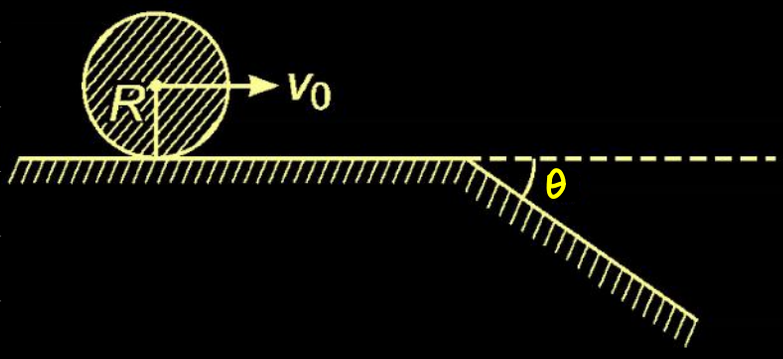
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Solution: a_{cm} of the system is $F/(m_1+m_2)$, this must also be the acceleration of the axle of the spool ($\because$ x coordinate of CM is same as that of the spool). Now α of the spool can be found using $\tau_{cm} = I_{cm}\alpha$ for the spool. Thus, $\alpha = FR/\left(\frac{1}{2}m_1R^2\right) = \dfrac{2F}{m_1R}$

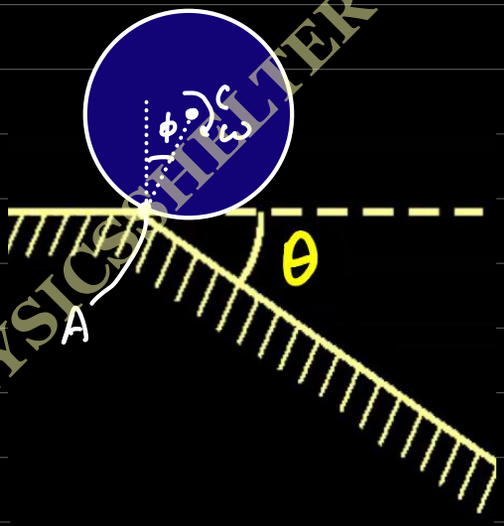
Now relative to the spool axle, K moves with $R\alpha$ i.e $2F/m_1$ thus from ground frame, $a_K = a_{cm} + R\alpha = \dfrac{F}{m_1+m_2} + \dfrac{2F}{m_1} = \dfrac{3m_1+2m_2}{m_1(m_1+m_2)}F$ Ans.

Now KE of the system must be equal to work done by this force Thus, KE : $F \times \frac{1}{2}a_K t^2 = \dfrac{F^2t^2}{2}\dfrac{(3m_1+2m_2)}{m_1(m_1+m_2)}$ Ans.

Example: Irodov 1.264:



A uniform solid cylinder of radius R rolls over a horizontal plane passing in to an incline plane of inclination θ. Find maximum permissible value of v_0 so that cylinder rolls on to the inclined section without any jump. The sliding is assumed to be absent.



Solution: As the cylinder reaches the edge, the edge serves as the instantaneous axis. As it pivots about A, its ω increases and so does the centrifugal force in the radial direction. At the same time, component of gravity along radial direction (AC) keeps on decreasing. Thus as the centre of the wheel keeps descending the normal reaction will keep decreasing. Thus least normal reaction occurs when $\phi = \theta$ after which pure rolling will take place on inclined surface. Now by energy conservation in limiting case:

$$\frac{1}{2} \times \frac{3}{2} mR^2 \left(\frac{v_0}{R}\right)^2 = -mgR(1-\cos\theta) + \frac{1}{2} \cdot \frac{3}{2} MR^2 \omega^2 \quad (1)$$

By centrifugal balance in the limiting case:

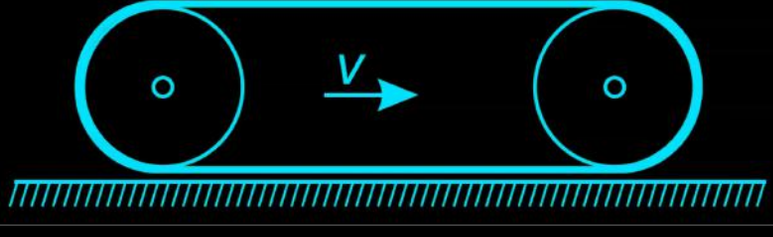
$$m\omega^2 R = mg\cos\theta \quad (2) \quad (\because N = 0).$$

Solving (1) & (2):

$$v = \sqrt{\frac{(7\cos\theta - 4)gR}{3}}$$

Example: Irodov 1.266:

Determine the kinetic energy of a vehicle's crawler belt of mass m if the vehicle is moving with a velocity v.



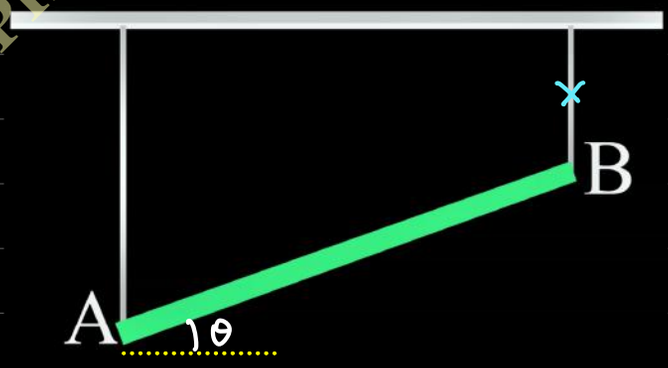
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Solution: The particles at the ground are at rest while those at the top are moving with $2v$. Thus from vehicle's frame, the entire belt is moving with a speed v (all particles). The KE in CM frame of the belt is $K_0 = \frac{1}{2}mv^2$. Now recall that total KE is given by $KE = K_0 + K_{cm}$, thus $KE = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 = mv^2$ Ans.

(∵ CM is also moving with speed v relative to ground).

ROTATIONAL CONSTRAINTS: Most problem involving constraints in rotational mechanics are based on the ideas of no slipping and Applications of derivatives using geometrical relations between various linear and angular coordinates. We will learn these ideas in upcoming examples.

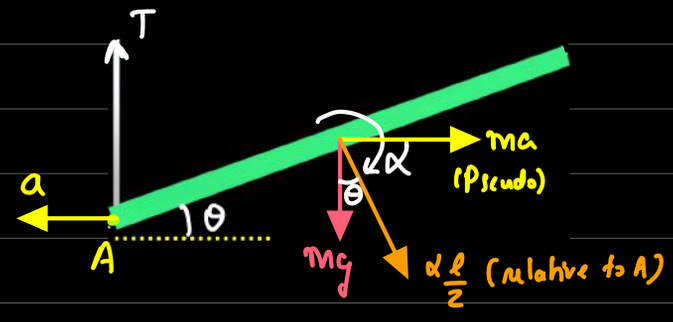
EXAMPLE: A rod of mass m and length ℓ hangs from two massless strings as shown. Now the right string is cut. What is the angular acceleration of the rod immediately afterwards.



Solution: We will use two ideas here,

① Since all the forces on rod are vertical, acceleration of its CM from ground frame must be vertical

② Since immediately after the cut, every particle has zero velocity and locus of A about the upper end of string is circle its centripetal acceleration is zero ∴ it can have only a horizontal acceleration (say a)



Now from frame of A, we must apply a pseudo force on CM as shown. Thus torque equation about A becomes:

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$$mg\frac{l}{2}\cos\theta + ma\frac{l}{2}\sin\theta = \frac{ml^2}{3}\alpha \quad (1)$$

Now using fact (1): $a - \alpha\frac{l}{2}\sin\theta = 0$ (2) ($\because a_{cm_x} = 0$)

Now solving (1) & (2):

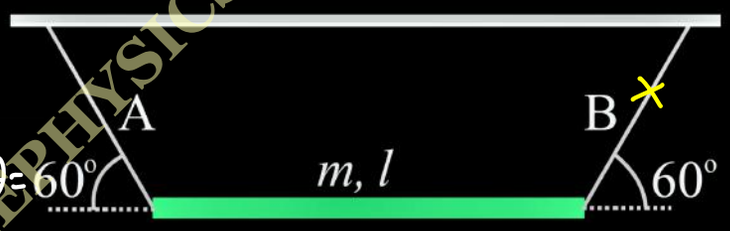
$$a = \frac{3g\sin\theta\cos\theta}{4-3\sin^2\theta}, \quad \alpha = \frac{6g\cos\theta}{l(4-3\sin^2\theta)} \text{ Ans.}$$

Think! How will you find T and acceleration of CM.

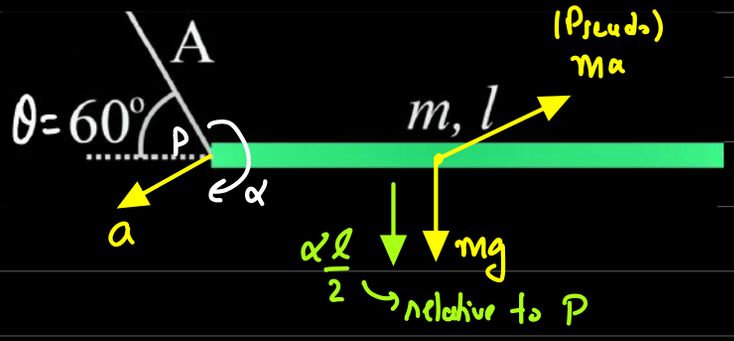
Solution: Making torque equation about CM: $T\frac{l}{2}\cos\theta = \frac{ml^2}{12}\alpha$

$\Rightarrow T = \frac{ml}{6}\alpha\cos\theta$ and $a_{cm} = \alpha\frac{l}{2}\cos\theta$ (α was already found).

EXAMPLE: A rod of mass m and length l hangs from two strings as shown in the figure. Now the string B is cut. Find angular acceleration of the rod just after cut.



Solution: Here there can be several ways of introducing the unknowns and making the equation, however we should look for solutions where we can do the problem in a small number of unknowns and equations. Here also like previous problem, acc^n of left end along string A is zero. Let its acc^n ⊥ to string be a. Thus we apply a pseudo force on the rod as shown. Now,



Torque equation: $mg\frac{l}{2} - ma\cos\theta\frac{l}{2} = \frac{ml^2}{3}\alpha$ (1)

Force equation ⊥ to string: $mg\cos\theta - ma = m\alpha\frac{l}{2}\cos\theta$ (2) (Avoided T!)

Now solving (1) & (2):

$$a = \frac{g\cos\theta}{4-3\cos^2\theta}, \quad \alpha = \frac{6g}{l}\left(\frac{1-\cos^2\theta}{4-3\cos^2\theta}\right) \text{ Ans.}$$

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Think: How will you find T and acceleration of CM?

Solution: Making torque equation about CM: $T \sin\theta \times \frac{\ell}{2} = \frac{m\ell^2}{12}\alpha$

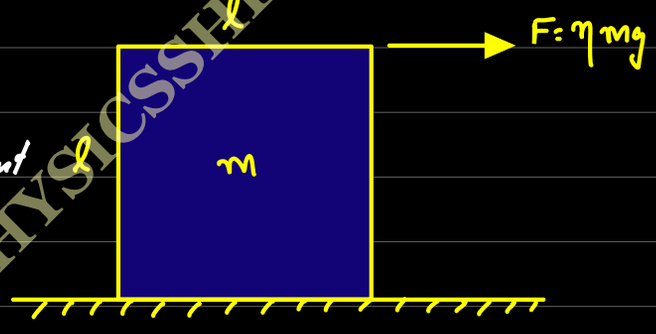
$$\Rightarrow \quad T = \frac{m\ell^2}{6}\operatorname{cosec}\theta \cdot \alpha \quad , \quad a_{CM_x} = a\sin\theta \quad , \quad a_{CM_y} = a\cos\theta + \alpha\frac{\ell}{2} .$$

→ Key learning: Try to assign acceleration variable to the constrained point

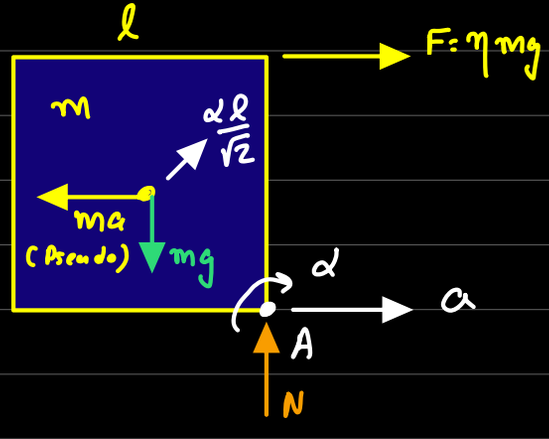
→ Try to make force equation in directions where unknown forces are avoided as much as possible

→ Try to make torque equation about points where torques of unknown forces are avoided as much as possible.

EXAMPLE: A force $F = \eta mg$ is applied on a cube. Find the angular acceleration of the cube assuming that F is sufficient to cause toppling.



Solution: Here bottom right corner is constrained to move along surface. So let us assume an acceleration a for the point A. Now writing torque equation about A (taking suitable pseudo force):



$$\eta mg\ell - ma\frac{\ell}{2} - mg\frac{\ell}{2} = \frac{2}{3}m\ell^2\alpha \quad (1)$$

Force equation along x: $\eta mg = m\left(a + \frac{\alpha\ell}{\sqrt{2}}\cos 45^\circ\right) = \left(a + \frac{\alpha\ell}{2}\right)m$ (2)

Solving (1) & (2): $a = \left(\frac{3+2\eta}{5}\right)g$, $\alpha = \frac{6}{5}(\eta - 1)\frac{g}{\ell}$

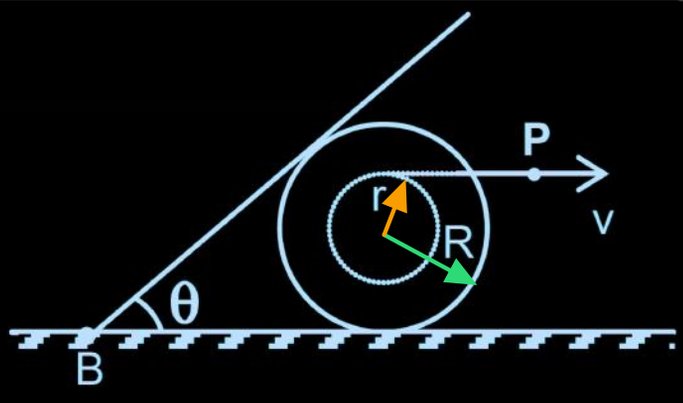
Think! How will you find normal reaction?

Solution: Vertical force equation gives $N - mg = m\left(\alpha\frac{\ell}{\sqrt{2}}\sin 45^\circ\right) = m\frac{\alpha\ell}{2} + mg$

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Example: Krotov: 1.16:

A bobbin rolls without slipping over a horizontal rough surface such that velocity v of the thread is horizontal. A board hinged at point B leans against the Bobbin as shown. Find the angular velocity of the board as a function of θ.



Solution:

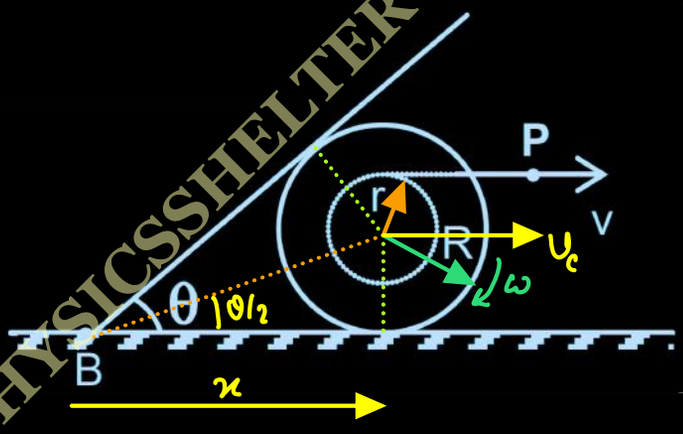
Consider the adjacent figure. Here,

$$v_c = \omega R = \frac{vR}{(R+r)} = \frac{dx}{dt} = \frac{d}{dt}\left(R\cot\left(\frac{\theta}{2}\right)\right)$$

$$\Rightarrow \frac{vR}{(R+r)} = -\frac{R}{2}\operatorname{cosec}^2(\theta/2)\cdot\frac{d\theta}{dt}$$

$\Rightarrow$ Angular velocity of board:

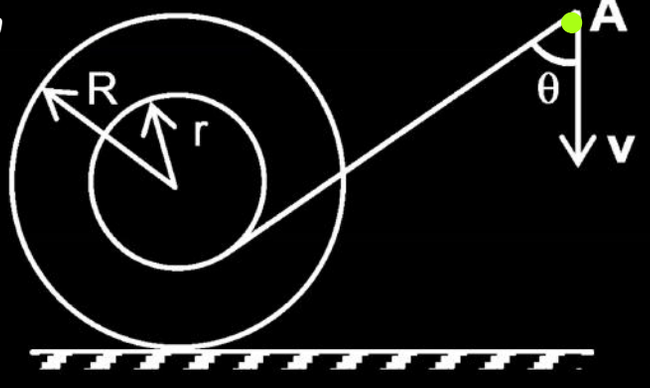
$$\frac{d\theta}{dt} = -2\sin^2(\theta/2)\frac{v}{(R+r)} \quad \text{Ans.}$$



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Example: Krotov: 1.27:

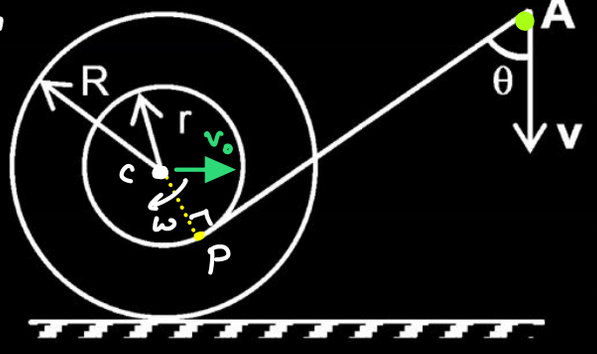
Free end of a thread wound on a bobbin of inner radius r and outer radius R is passed around a nail A hammered in to the wall. The thread is pulled with a constant velocity v. Find the velocity v_0 of the centre of the bobbin when the thread forms an angle θ with vertical as shown. Assume pure rolling.



Solution: Let angular velocity of Bobbin be ω then $\omega = v_0/R$. Also speed of P along the string:

$$v_p = v_0\sin\theta - \omega r = v_0\sin\theta - \frac{v_0}{R}r = v$$

$(\because \vec{v}_p = \vec{v}_c + \vec{v}_{pc})$, thus:



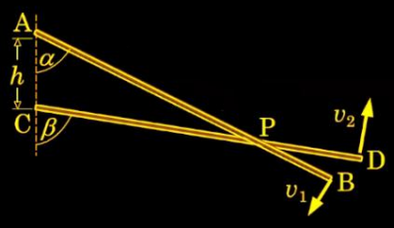
$$v_0\left[\frac{-r}{R}+\sin\theta\right]=v \Rightarrow v_0 = v/\left[\frac{-r}{R}+\sin\theta\right] \quad \text{Ans.}$$

Think! What is $d\theta/dt$ if distance $AP = l$?

Solution: $d\theta/dt = v_{P_1}/l = v_0\omega\theta/l$.

→ Once we know $d\theta/dt$ we can also find a_c and α by differentiating the velocity expression. We could also find pulling force if we knew moment of inertia (using $\alpha = Fv/I_{eff}\omega$).

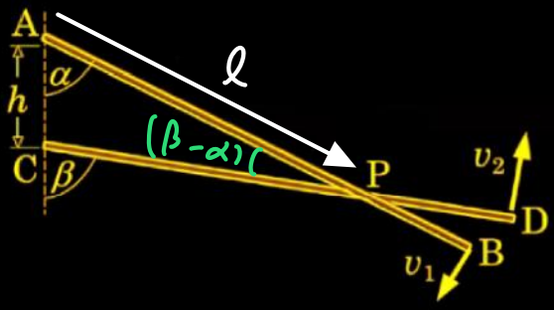
EXAMPLE: PATHFINDER: CHECK-1 :



Two identical thin rigid rods AB and CD each of length L touching each other at point P rotate in a plane about their stationary ends A and C that are a distance h apart. Velocities of their ends B and D have constant moduli v_1 and v_2 respectively. Find component of velocity of their point of intersection P along the rod AB when the rods make angles α and β with the line AC.

Solution: By sine law:

$$\frac{AP}{\sin\beta} = \frac{h}{\sin(\beta-\alpha)} \Rightarrow AP = \frac{h\sin\beta}{\sin(\beta-\alpha)} = l$$

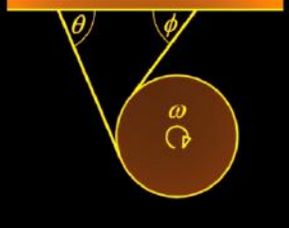


$$\Rightarrow \frac{dl}{dt} = \frac{h}{\sin^2(\beta-\alpha)}\left[\begin{array}{l}\sin(\beta-\alpha)\cos\beta\,\dot{\beta} \\ -\sin\beta\cos(\beta-\alpha)(\dot{\beta}-\dot{\alpha})\end{array}\right]$$

Also $\dot{\beta} = v_2/L$, $\dot{\alpha} = -v_1/L$

$$\Rightarrow \frac{dl}{dt} = \frac{-h}{L\sin^2(\beta-\alpha)}\left[v_2\sin(\beta-\alpha)\cos\beta - (v_1+v_2)\sin\beta\cos(\beta-\alpha)\right] \quad \text{Ans.}$$

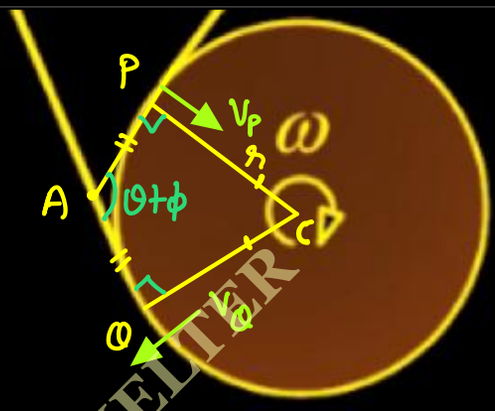
EXAMPLE: PATHFINDER: CHECK-2:



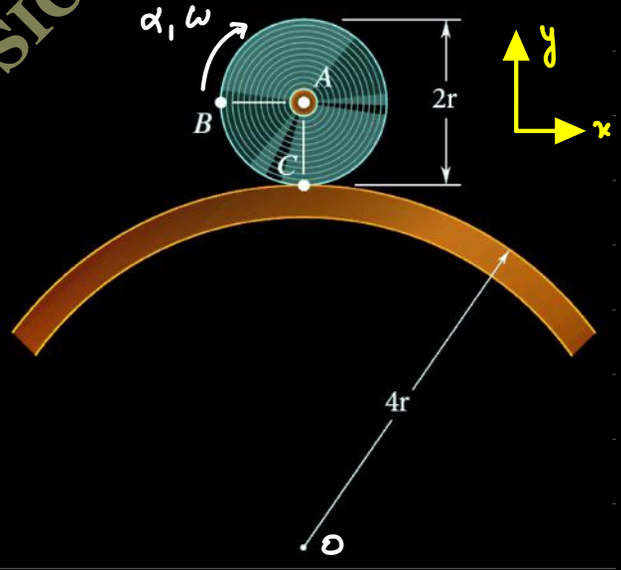
Free ends of two inextensible cords wrapped on a massive disc of radius $r = 10\sqrt{3}$ cm are affixed to the ceiling at two different points. Both the cords and the disc are in the same vertical plane. At an instant after the disc is released, angles made by the cord with the ceiling are $\theta = 67°$ and $\phi = 53°$ and angular velocity of the disc is $\omega = 5.0$ rad/s. Find magnitude of velocity of the centre of the disc at this instant.

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Solution: The contact points of the disc and thread must have same velocities. Also since threads are inextensible, the contact points must have velocities perpendicular to the thread. This gives us an easy method to locate instantaneous axis of the disc.
This process is geometrically shown in the adjacent figure. Here A is the instantaneous axis thus $V_C = \omega \times AC = \dfrac{\omega R}{\sin\left(\frac{\theta+\phi}{2}\right)} = 1\ m/s$



EXAMPLE: A wheel rolls without slipping on a fixed cylinder as shown. At the shown instant angular velocity of the wheel is ω clockwise and its angular acceleration is α clockwise. Find accelerations of (a) A (b) B and (c) C.



Solution: A is moving in a circle of radius $5R$, C is its instantaneous axis. Speed of A is ωR accordingly its centripetal acceleration towards O is $(\omega R)^2/5R$ also its tangential acceleration is $R\alpha$ ($\because$ pure rolling). Thus its net acceleration is $\vec{a}_A = \dfrac{\omega^2 R}{5}(-\hat{j}) + R\alpha\,\hat{i}$ Ans.

Now from frame of A, acceleration of B is $\alpha R\,\hat{j} + \omega^2 R\,\hat{i}$
Thus, $\vec{a}_B = \vec{a}_A + \vec{a}_{BA} = (\omega^2 R + R\alpha)\,\hat{i} + \left(\alpha R - \dfrac{\omega^2 R}{5}\right)\hat{j}$ Ans.

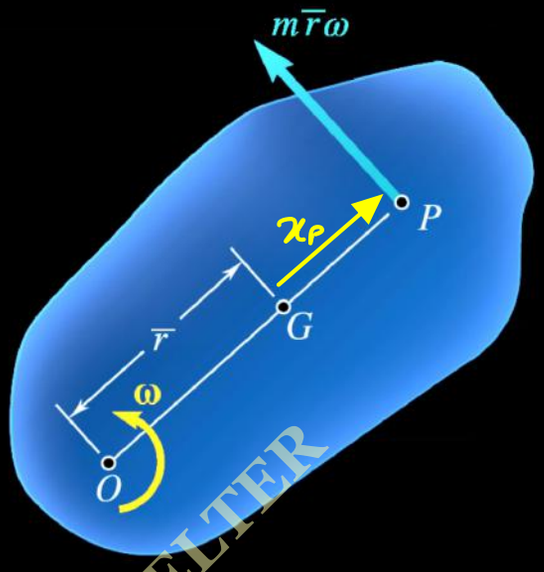
Now C has no tangential acceleration. While from frame of A, its upward acceleration is $\omega^2 R\,\hat{j}$ thus its net acceleration:

$$\vec{a}_C = \omega^2 R\,\hat{j} - \frac{\omega^2 R}{5}\hat{j} = \frac{4}{5}\omega^2 R\,\hat{j}$$ Ans.

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CENTRE OF PERCUSSION:

The point on line joining a pivot to the centre of mass of a rigid body where entire linear momentum can be assumed to be concentrated to get the angular momentum of the rigid body about the pivot is called centre of percussion. Now we will find the location of centre of percussion using this definition. Let x_p be the distance of centre of percussion from the centre of mass G. Further $\bar{r}$ is location of CM wrt pivot. Now if the body is rotating with angular velocity ω, its linear momentum is $p = m\bar{r}\omega$ and if k is its centroidal radius of gyration, then angular momentum is $L = m(k^2 + \bar{r}^2)\omega$



Now by definition of centre of percussion,

$$m\bar{r}\omega(\bar{r} + x_p) = m(k^2 + \bar{r}^2)\omega, \quad \text{Now}$$

rearranging and cancelling terms, we get $\bar{r}^2 + \bar{r}x_p = \bar{r}^2 + k^2$

$\Rightarrow x_p = k^2/\bar{r}$. To memorize this formula we will write it in the following form:

$$\boxed{x_p = \frac{k_{cm}^2}{r_{cm}}} \quad (1)$$

k_{cm}: Centroidal radius of gyration
r_{cm}: location of cm from pivot.

→ When a force or impulse acts on a body at centre of percussion perpendicular to line OP it creates no impulse at the pivot.

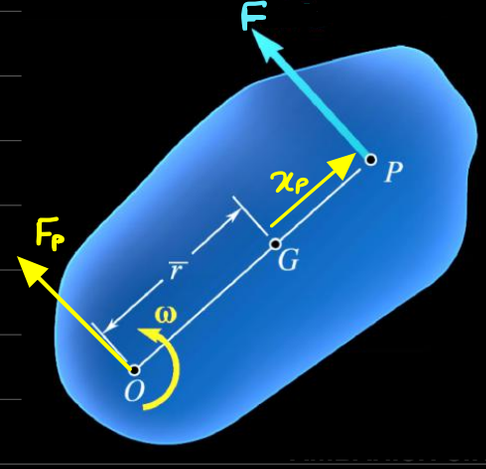
This can be shown as follows.

Consider the adjacent figure:

$$\alpha = \frac{F(\bar{r} + x_p)}{m(k^2 + \bar{r}^2)}$$

$$\Rightarrow a_{cm} = \bar{r}\alpha = \frac{F\bar{r}(\bar{r} + x_p)}{m(k^2 + \bar{r}^2)} = \frac{F + F_p}{m} \quad (1)$$

where F_p is the force from pivot $\perp$ to OP.



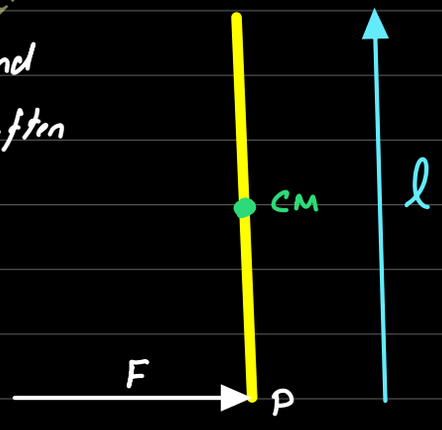
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From (1): $\frac{F[\bar{x}^2 + \bar{x}x_p]}{k^2 + \bar{x}^2} = F + F_p$, Now $x_p = k^2/\bar{x}$

$\Rightarrow \frac{F[\bar{x}^2 + \bar{x}\cdot k^2/\bar{x}]}{k^2 + \bar{x}^2} = F + F_p \Rightarrow F_p = 0.$

→ Velocity of the pivot is zero. When an unpivoted body is hit somewhere, there will exist some point on it where the velocity is zero, this point can be easily located using centre of percussion idea. (This point is instantaneous axis immediately after the hit). To locate this point we can rewrite this equation as $x_{cm} = \frac{k^2}{x_p}$

EXAMPLE: A uniform thin rod of mass m and length l lies on a frictionless table. Now a force F is applied very briefly at its one end as shown. Locate the instantaneous axis just after application of the force for a brief time.



Solution: Instantaneous axis is the point where the velocity is zero, thus it would not have mattered even if the body were pivoted there because it will not offer any force. Thus we are looking for location of the pivot effectively. We will do it using two methods:

Method I: Centre of percussion: $x_{cm} = \frac{k_{cm}^2}{x_p}$, here the point P is the centre of percussion. Thus, $x_{cm} = \frac{l^2/12}{(l/2)} = l/6.$

Thus effective pivot is at a distance $l/6$ above the CM (away from P).

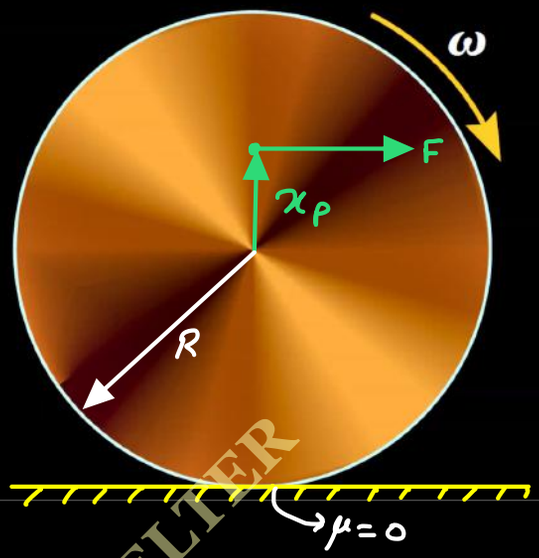
Method II: Basic approach: $v_{cm} = \frac{F\,dt}{m}$, $\omega = \frac{(F\,dt)\,l/2}{I_{cm}} = \frac{(F\,dt)\,l}{\left(\frac{ml^2}{12}\right).2}$

$\Rightarrow \omega = \frac{6F\,dt}{ml}$, Now distance of instantaneous axis:

$x_{cm} = \frac{v_{cm}}{\omega} = l/6$ which is the same result that we obtained above.

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Centre Of Percussion For Pure Rolling Wheels (Sweet Spot):



The point above the centre of a wheel where a horizontal force may be applied so that the wheel may roll without slipping even on a frictionless surface is called the sweet spot (or centre of percussion of the wheel assuming bottommost point (IC) as the pivot. We will derive its location from basics and prove that it comes out to be the same formula as we derived for centre of percussion.

Let I_{cm} be the moment of inertia of the wheel about CM and let x_p be the required height above CM to achieve pure rolling. Then:

$a_{cm} = F/m$ and $\alpha = \dfrac{F x_p}{I_{cm}}$ ($\because \tau_{cm} = I_{cm}\alpha$).

Now for pure rolling, $a_{cm} = \alpha R$ thus: $\dfrac{F}{m} = \dfrac{F x_p}{m k_{cm}^2} \times R$

$\Rightarrow x_p = \dfrac{k_{cm}^2}{R}$, this is exactly the same result we would obtain using direct formula $x_p = k_{cm}^2 / r_{cm}$. Here r_{cm} is R (i.e distance of instantaneous axis from CM).

Thus the height of sweet spot is given as:

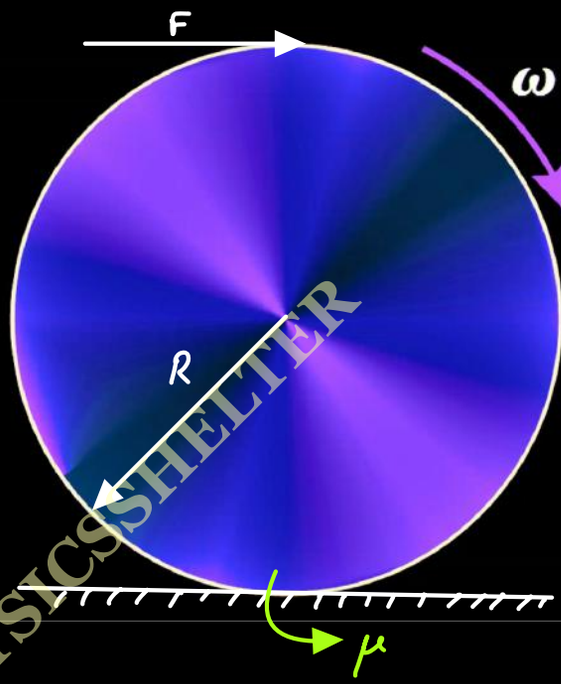
$$\boxed{x_p = \frac{k_{cm}^2}{R}}$$

(2) If the force were applied at a height greater than x_p, it would increase the angular acceleration, while maintaining the same value of a_{cm}. Thus it would cause backward slipping on a smooth surface (as if the surface were rough, friction will act forward). If the force were applied at a height less than x_p, it would reduce the value of α while keeping same value of a_{cm} thus causing a forward slipping on smooth surface (or backward friction if surface were rough).

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From formula ②, centre of percussion can be readily recognized e.g.: Ring: $x_p = R$, Disc: $x_p = R/2$, Solid Sphere: $x_p = \frac{2}{5}R$ etc.

How To Decide Whether A Wheel Will Roll Or Slip:

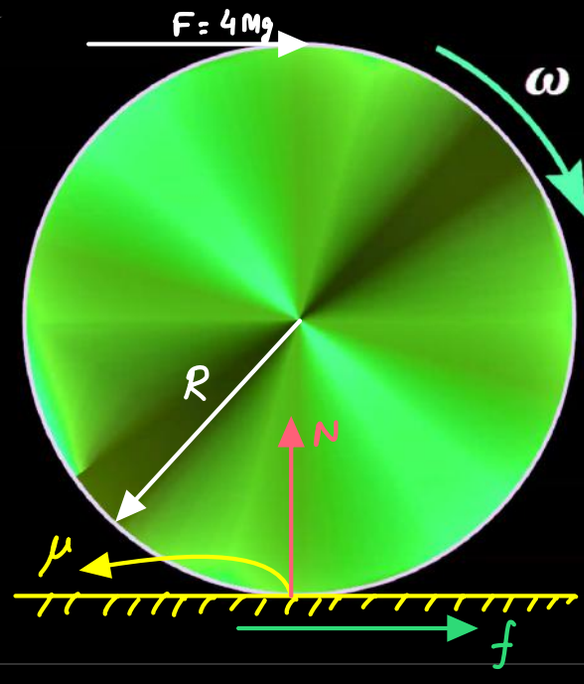


Just as in case of block on block problems, in rolling problems we are not initially sure about whether or not friction is sufficient for pure rolling. Here to find out the case we follow the following procedure:

1. Assume pure rolling
2. Find the required value of friction for pure rolling using force or torque equations (Call it f_n)
3. Compare f_n with limiting value of friction ($f_l = \mu N$) if $f_n < \mu N$ our assumption is correct and if $f_n > \mu N$ our assumption is incorrect and we need to solve the problem again assuming kinetic friction.

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Example: A force of $4Mg$ is applied to the topmost point of a disc which stands on a surface with $\mu = 0.5$ as shown in the figure. Will the wheel slip on roll without slipping? Also find its angular acceleration and frictional force.



Solution: Using $\tau_A = I_A \alpha$,

$$4Mg \times 2R = \frac{3}{2} MR^2 \alpha$$

$$\Rightarrow R\alpha = 16g/3 = a_{cm} \quad (\text{Assuming pure rolling}).$$

Now $f + 4Mg = Ma_{cm} = \frac{16Mg}{3} \Rightarrow f = \frac{4Mg}{3}$.

This is the required friction for pure rolling. But limiting friction, $f_L = \mu N = \mu Mg$. Thus required value for rolling exceeds the limiting friction, ∴ the wheel will definitely slip. We know that friction acts forward ∵ height of force is more than centre of percussion.

Now for the slipping wheel, $\alpha = \tau_{cm}/I_{cm} = \dfrac{4Mg R - 0.5 Mg R}{(\frac{1}{2}MR^2)}$

$\Rightarrow \alpha = 7g/R$ Ans., $a_{cm} = F_{net}/M = (4Mg + 0.5Mg)/M$

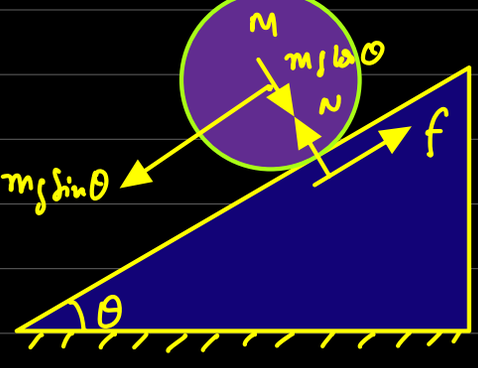
$\Rightarrow a_{cm} = 4.5g$ Ans.

→ Here we knew the direction of friction beforehand using concept of sweet spot, however sometimes we may not know it. In case of pure rolling you may assume friction to be in any direction. If it comes out negative, it is in opposite direction to the assumed direction.

→ Just like in block on block problems if $\mu_s = \mu_k = \mu$ we could also start with kinetic assumption, then we will have to check the direction of relative acceleration at the contact point should be opposite to the assumed direction of friction.

Minimum Coefficient Of Friction For Pure Rolling On An Incline:

Consider a wheel having moment of inertia I_{cm} about centroidal axis and I_A about instantaneous axis (pure rolling). Then:



$\tau_{cm} = I_{cm}\alpha \Rightarrow fR = I_{cm}\alpha$ ①

also $\tau_A = I_A\alpha \Rightarrow Mg\sin\theta R = I_A\alpha$ ②

①/② $\Rightarrow$

$$\frac{f}{Mg\sin\theta} = \frac{I_{cm}}{I_A}, \text{ also } N = Mg\cos\theta$$

$$\Rightarrow Mg\sin\theta = N\tan\theta.$$

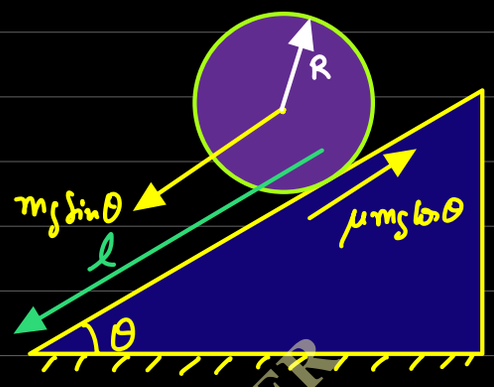
$$\Rightarrow \frac{f}{N\tan\theta} = \frac{I_{cm}}{I_A} \Rightarrow \frac{f}{N} = \frac{I_{cm}}{I_A}\tan\theta \leq \mu$$

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Thus for pure rolling of a wheel, $\mu \geq \frac{I_{cm}}{I_A} \tan\theta$

Work Done By Friction In Slipping:

Suppose a wheel slips on an incline through a distance l starting from rest. To find the work done by friction one can't write $W_f = -\mu mg\cos\theta \times l$. That would be a blunder. Recall that work is calculated considering displacement of particle on which friction is applied and here continuously different particles are coming in contact with the incline, therefore we must integrate the power developed by friction. Thus if instantaneous speed of the wheel is v and its angular velocity is ω, if the wheel is slipping downward, the Power developed by friction is:



$$P_f = -\mu mg\cos\theta\ [v - \omega R] = -\mu mg\cos\theta\, v + \mu mg\cos\theta\, \omega R.$$

We can think of the power developed by friction as superposition of two terms, the first term representing power if there were no ω and the body were only translating and the second term represents power if the wheel were only rotating with centre of wheel at rest. Thus the second term represents rate of change of KE if the wheel was undergoing fixed axis rotation about centroidal axis. Thus:

$$P_f = P_{Trans} + P_{rot} \quad \text{where} \quad P_{trans} = -\mu mg\cos\theta\, v$$
$$P_{rot} = \mu mgR\,\omega\cos\theta.$$

According $W_f = W_{trans} + W_{rot}$

Now $W_{trans} = \int -\mu mg\cos\theta\, v\, dt = -\mu mgl\cos\theta$

and $W_{rot} = \int \left(\frac{dKE}{dt}\right)_{rot} dt = \Delta KE_{rot} = \frac{1}{2} I_{cm}(\omega^2 - \omega_0^2)$

Where ω and ω_0 are final and initial angular velocities.

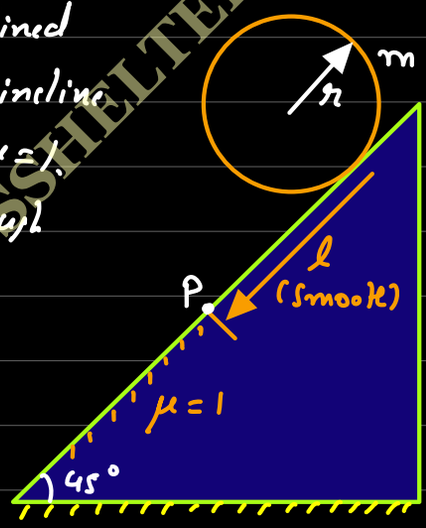
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Alternatively we could find work done by friction using work energy theorem i.e $ME_f - ME_i = W_f$.

→ In pure rolling, net work done by friction will be zero (in pure rolling, friction is a static friction pair and work on incline is zero from its own frame ∴ work on wheel is also zero in that frame).

→ If part of an incline is smooth and another part rough, it is possible that after slipping for some distance, pure rolling might start so one has to carefully consider the position where this happens.

EXAMPLE: A ring begins to slip on an inclined plane of angle 45°. Initial l distance of the incline is smooth and then the rough surface has $\mu = 1$. How much time does the ring spend on rough patch before beginning pure rolling?



Solution: Speed at the beginning of rough patch, $v_p = \sqrt{2gl\sin 45°}$, $\omega_p = 0$.

After this $\Delta v = g(\sin\theta - \mu\cos\theta) = 0$ (Until slipping continues).

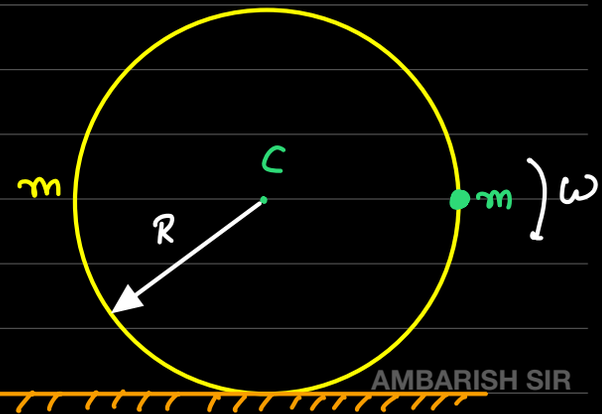
$\alpha = \mu mg\cos\theta R / I_{cm} = g\cos\theta/R$

Now for pure rolling, $v = \omega R = \alpha R t = \sqrt{\sqrt{2}gl} = \frac{g}{\sqrt{2}} \times t$

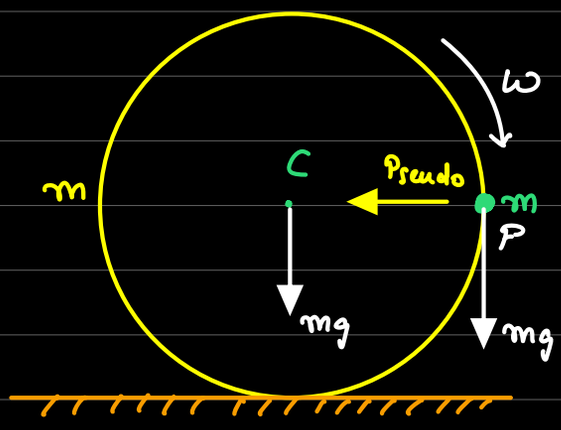
$\Rightarrow t = \sqrt{\frac{2\sqrt{2}l}{g}}$ ans.

Now if we want we can use work energy theorem to work out loss in mechanical energy (-ve of work done by friction).

EXAMPLE: A ring of mass m has a bead of same mass welded to its periphery as shown. At the shown instant, the ring has angular velocity ω. The surface is smooth. What is the angular acceleration of the ring?



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Solution: Here centre of mass must have zero horizontal acceleration because there is no external force in x direction. Note that bottommost point is not the instantaneous axis and centre of the ring is not the centre of mass of the system. So we need to be careful. Let us work from centre of the ring frame. Let its horizontal acceleration be a. Relative to C, horizontal acceleration of P is $\omega^2 R$ along $-x$.

Thus using $\Sigma m_i a_{x_i} = 0$, $\quad m a + m(a - \omega^2 R) = 0 \Rightarrow a = \omega^2 R/2$.

Now the good thing is that since C has a horizontal acceleration, pseudo force on the bead will pass through C and hence not generate any torque. Now using $\tau = I\alpha$ about C: $mgR = (mR^2 + mR^2)\alpha$

$\Rightarrow \alpha = g/2R$ Ans.

→ Had m been above or below the level of C, α would have depended on ω.

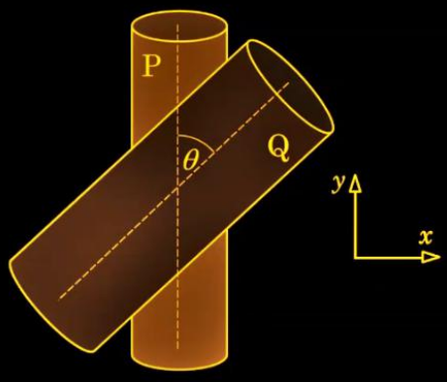
Think! What is the force applied by the ring on the bead?

Solution: Acceleration of bead is $(g/2)(-\hat{j}) - (\omega^2 R/2)\hat{i} = mg(-\hat{j}) + \vec{F}_R$

$\Rightarrow \vec{F}_R = mg/2\,\hat{j} - \omega^2 R/2\,\hat{i}$ Ans.

EXAMPLE: PATHFINDER: MCQ13:

A cylinder P of radius r_P is being rotated at a constant angular velocity $\omega_P \hat{j}$ with the help of a motor about its axis that is fixed. Another cylinder Q of radius r_Q free to rotate about its axis that is also fixed is touched with and pressed on P making an angle θ between their axes. Soon after the cylinders are pressed against each other, a steady state is reached and the cylinder Q acquires a constant angular velocity. What can you conclude when the steady state is reached?



(a) Angular velocity of cylinder Q is $\vec{\omega}_Q = -\dfrac{\omega_P r_P}{r_Q \cos\theta}(\sin\theta\hat{i} + \cos\theta\hat{j})$.

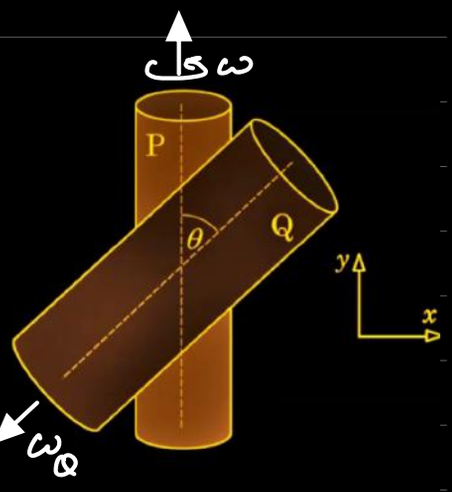
(b) Angular velocity of cylinder Q is $\vec{\omega}_Q = -\dfrac{\omega_P r_P \cos\theta}{r_Q}(\sin\theta\hat{i} + \cos\theta\hat{j})$.

(c) Unit vector of frictional force on cylinder Q is $(\sin\theta\hat{i} + \cos\theta\hat{j})$.

(d) Frictional force on each cylinder becomes vanishingly small.

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Solution: The contact point of P with cylinder Q has a velocity $\omega_P r_P \hat{i}$ which has a component $\omega_P r_P \cos\theta$ perpendicular to axis of Q. Due to rubbing between cylinders, this component will be equal for particles of both cylinders after some time. The other component $\omega_P r_P \sin\theta$ can't become equal due to the axis constraint of Q. Thus $\omega_Q r_Q = \omega_P r_P \cos\theta \Rightarrow \omega_Q = \dfrac{\omega_P r_P \cos\theta}{r_Q}$



Thus, $\vec{\omega}_Q = -\dfrac{\omega_P r_P \cos\theta}{r_Q}\left[\sin\theta\,\hat{i} + \cos\theta\,\hat{j}\right]$ option (b) is correct

Now cylinder Q will be rubbed along the component $\omega_P r_P \sin\theta$ along the axis of Q. Thus friction on Q acts along $\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$ thus option (c) is also correct. Correct options: b and c.

EXAMPLE: PATHFINDER : MCQ 14:



Figure-I Figure-II

A uniform square plate is placed on a horizontal floor. When it is given an angular velocity ω about a vertical axis through one of its corners as shown in figure-I, it takes time t_1 to come to a complete stop. Now the same square plate is given the same angular velocity to rotate about another vertical axis through its centre as shown in figure-II. How long will it take to come to a complete stop now?

(a) $t_1/2$ (b) $2t_1/3$
(c) $3t_1/2$ (d) $2t_1$

Solution: This is a problem in scaling wherein we need to figure out the factor by which the quantity τ/I is scaled. Figure II can be considered to be made of four half scale models of figure I (4 squares making bigger square). For each small square, mass is 1/4 times and linear dimension is $\frac{1}{2}$ times. Thus moment of inertia is 1/16 times. Now pressure per unit area is same but area is 1/4 times and distance from axis is $\frac{1}{2}$ times. Thus frictional torque becomes 1/8 times. Thus overall, τ/I becomes (1/8)/(1/16) i.e 2 times, so angular acceleration is double accordingly stopping time must get halved. Thus correct option is (a) Ans.

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